

LAB-2310

Your practical guide to troubleshooting for Webex Contact Center

Anuj Bhatia, D.K. Singh

Copyright © 2025 Cisco

Table of contents

1. Lab	3
1.1 Overview	3
1.2 Lab 1 - Configure and Debug Tenant Call Flows	4
1.3 Lab 2 - Seamless Call Transfer & Preserving flow Variables	28
1.4 Lab 3 - Set Up Outdial Calls in WxCC	36
1.5 Task 4 - Exploring WebRTC data and statistics	53
1.6 Conclusion	59
2. How to use this guide	0
2.1 Setting up and using the tool	0
2.2 Styling and Advanced Features	0

1. Lab

1.1 Overview

1.1.1 Learning Objectives

This lab session is an intermediate-level session for engineers and administrators interested in the voice channel and its related troubleshooting techniques in a Webex Contact Center Solution. The session is divided into three exercises that cover following topics:

- **Configure and manage inbound call flows:** Participants will learn to set up inbound call flows and call distribution strategies.
- **Debug and troubleshoot common issues:** The lab will focus on using tools like flow debuggers to identify and fix configuration errors, whether in inbound flows or outbound dialing setups.
- **Handle call transfers and data preservation:** Participants will gain experience in ensuring a seamless call transfer process that maintains crucial data (flow variables) for agents.
- **Analyze agent performance:** The exercises will allow participants to use call reports to understand and analyze agent behavior, helping them identify the root cause of specific call events like rejections or "Re-Route on No Answer" (RONA) events.
- **Set up outdial capabilities:** The lab will cover the process of configuring a system to enable agents to make outbound calls.

Essentially, the lab is designed to give participants a holistic view of the call center solution, from initial setup and configuration to troubleshooting complex issues and analyzing performance related to the voice channel.

1.1.2 Disclaimer

Although the lab design and configuration examples could be used as a reference, for design related questions please contact your representative at Cisco, or a Cisco partner.

1.1.3 Lab Access

This lab leverages Webex Contact Center cloud environment.

1.2 Lab 1 - Configure and Debug Tenant Call Flows

Please use the following credentials to connect to Control Hub and configure Webex Contact Center:

Control Hub URL	https://admin.webex.com
Username	labuser ID @wx1.wbx.ai (where ID is your assigned pod number; this ID will be provided by your proctor)
Password	webexONE1!

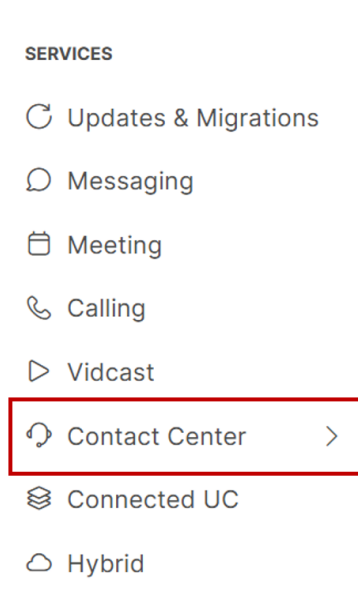
1.2.1 Objective

The objective of this self-paced lab is to provide participants with hands-on experience in configuring and debugging tenant call flows, as well as analyzing agent performance through call reports. This exercise has been divided into three sections

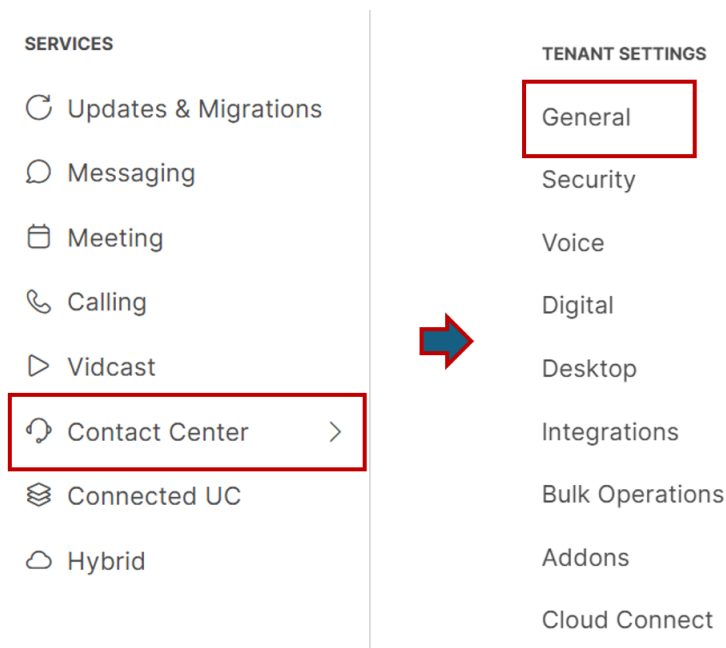
- **Section 1:** Discover Tenant Call Flow Configuration: This section aims to familiarize participants with the existing call flow configurations within a tenant environment.
- **Section 2:** Configure and Debug an Inbound Flow: In this part, participants will set up an inbound flow with a whisper announcement and a "Team" and "Longest Available" distribution strategy. They will use the flow debugger to identify and resolve common misconfigurations encountered during the setup process.
- **Section 3:** Analyze Agent Performance with Call Reports: After creating the inbound flow and presenting calls to a test agent, participants will intentionally trigger specific events. They will reject one call and allow a second to result in a Re-Route on No Answer (RONA) event. This will enable them to explore and analyze call reports to understand agent behavior and pinpoint the root cause of the rejected and RONA calls.

1.2.2 Section 1 : Discover Tenant Call Flow Configuration

- Open your web browser and navigate to <https://admin.webex.com>
- Log in using the provided credentials.
- In Control Hub, locate Services and click on Contact Center.



- In the Contact Center navigation pane on the left side, scroll down to the Tenant Settings section and click on General.



- Under the General settings, locate the Telephony type. Confirm that it states: **Webex Calling**.

Service Details	Country of operation	United States of America
	Tenant Timezone	(GMT-05:00) America/New_York
	Changes to this field will only affect your voice channels and will not impact your digital cha	
	Platform details	Webex Contact Center
	Digital channel	Webex Connect
	Voice media platform	Real Time Media Service
	Telephony	Webex Calling

ⓘ This PSTN Connection type was selected during first-time setup. To learn abc

- In the Contact Center navigation pane, under the Customer Experience section, select Channels and search for the entry point named: **Entry Point-1**.

Contact Center

Overview

CUSTOMER EXPERIENCE

Channels

Queues

Business Hours

Audio Files

Flows

Call Recording Schedules

Channels

Channels

Settings

Search by name or number

Active

3 Channels

Channel

Description

Channel type

Entry Point-1

Entry point created by system

Inbound Telephony

Outdial Entry Point-1

Outdial Entry Point created by system

Outbound Telephony

- Within the details for **Entry Point-1**, locate the Phone Number section and note the Webex Calling location and the Support Number.
- Webex Calling location: [RTP Location 8]
- Support Number: [+19842906065]

Phone numbers

This is the phone number customers will call to reach your business. You can select multiple from the numbers you have configured under \

Number

Webex Calling location

Support number

1

RTP Location 8

+19842906061

Add more

- Go back to the main menu of Control Hub.
- Under the Management section, select Locations.
- In the Locations search bar, enter the Webex Calling location name you discovered in the previous step.

MANAGEMENT

Users

Groups

Locations

Workspaces

Devices

Apps

Account

Security

Organization Settings

Locations

List

Map

Beta

View location(s) with pending action items to resume your calling service in 1 or more locat

Search by name or addr...

Filter

1 location

Location name

RTP Location 8

- Select the discovered location.
- In the **PSTN tab**, under the **PSTN Configuration** section, the identified connection is **Cisco Calling Plan**.



RTP Location 8

Location ID: f226bead-d0ca-4ba2-a21f-55f5577663b9

33 users • 0 workspaces

Overview Floors Calling **PSTN**

PSTN Configuration

PSTN connection ⓘ

Cisco Calling Plan [Manage](#)

Main number ⓘ

+19842906065

- Click on the **Manage** section and it provides all the details of the **connection type**

Connection Type

Cisco Calling Plans (US) ⓘ

Subscription
trial

Emergency

7100 Kit

Morrisville, NC 27560

US

Contract Information

[Edit](#)

WxCC Event Tenant

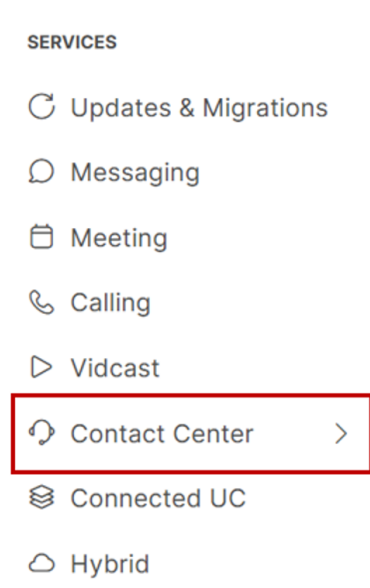
mdanylch@cisco.com

Cisco-provided PSTN provides a bundled Cisco solution that simplifies your cloud calling experience with easy PSTN ordering and full support from Cisco and our Partners.

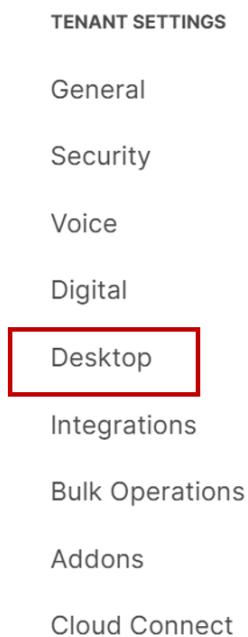
- Based on the observed calling connection (e.g., Cisco Calling Plan) and the information presented in the call flow slides, this configuration matches Flow Type 1.

1.2.3 Section 2 : Configure and Debug an Inbound Flow

- The steps below covers the process for handling an incoming call, ensuring all necessary settings are in place for the agent to receive the call.
- First, to enable call handling options for the agent, navigate to "Contact Center" in the Control Hub.



- Go to "Tenant Settings" and select "Desktop"



- Confirm that
- "**End Call**" and "**End Consult**" is enabled.
- "Auto Wrap-up interval" is set to **600 seconds**.
- "Telephony RONA timeout" is set to **12 seconds**.

Agent Experience	Allow desktop inactivity after given duration.	
Inactivity timeout	☒	
Inactive duration *	3600	Minutes
Auto Wrapup Interval *	600	Seconds

Voice Features	Enable Force Default DN	☐
Enable End Call	☒	
Enable End Consult	☒	

RONA Timeouts	Telephony *	12	Seconds
	Chat *	32	Seconds
	Email *	32	Seconds
	Social *	32	Seconds

- Navigate to "Desktop Experience" and select "Desktop Profiles."

DESKTOP EXPERIENCE

AI Assistant

Multimedia Profiles

Outdial ANI

Desktop Layouts

Dial Plans

Address Books

Desktop Profiles

Idle/Wrap-up Codes

- Choose the **WebexOne_AgentProfile** and verify that the "Voice Channel" options are configured as shown in the following screenshot.

Agent-Profile

ID: 298f8f79-85d7-4b86-9e6c-b271c0ddccecc • Last Modified: July 30, 2025 15:54 PM

General

Idle/Wrap-up Codes

Collaboration

Dial Plans

Voice Channel options

Agent Statistics

Voice Channel options

One Voice option must always be checked.

☒ Agent DN

☒ Extension

☒ Desktop

Validation for Agent DN

☒ Unrestricted (Allow any value)

☐ Provisioned DN (Restrict login DN to provisioned agent DN)

☐ Validate using Dial Plans (Select from list)

 Note

Depending on the customer's business needs, the desired option can be selected. For example, if a customer wants all of their agents to use WebRTC and no other endpoints, only the 'Desktop' option can be checked. Agents will then only see the desktop option when logging into the desktop.

- To confirm that the agent is mapped to this profile, go to the "Contact Center users" section in "User management."

USER MANAGEMENT

Sites

Skill Definitions

Skill Profiles

Teams

User Profiles

Contact Center Users

- Search for your agent **labuserid@wx1.wbx.ai** and, in their agent settings, ensure that the "desktop profile" is mapped to **WebexOne_AgentProfile**.

Agent Settings	Site	Site-1
	Teams	TeamVoice
		1 Teams Clear All
	Desktop Profile	Anuj_Outdial_Profile
	Multimedia Profile	Default_Multimedia_Profile

- Now, let's start building the entry point for the external call. To do this, navigate to "Customer Experience" and select "Channels."

CUSTOMER EXPERIENCE

Channels

Queues

Business Hours

Audio Files

Flows

AI Agents

Call Recording Schedules

- Click "Create a channel."

- In the "Channel Creation Wizard," provide the following details:
- Name: **Provide a descriptive name for your channel**
- Channel Type: **Inbound Telephony**

Entry point	Name *	Enter a name
	Description	Enter a short description
	Channel type *	Inbound Telephony

- Service Level Threshold: **30 seconds**
- Timezone: **America/New York**
- Routing flow: **Select Webexone_Flow_[num]** (Num is the assigned assignee's number)
- Music on hold: **defaultmusic_on_hold**
- Version label: **Latest**

Entry point settings

Service level threshold ⓘ *	00 Seconds
Timezone (Business hours only) *	America/New_York
Routing flow	MS_Dynamics
Music on hold *	defaultmusic_on_hold.w...
Version label *	Latest

The flow version may impact the caller's experience and the variables available to override.

Override flow settings



No overrides available

This flow doesn't have any variables set up for overrides.

- In the "Phone numbers" section, select the following:
- Number Webex Calling location: **RTP Location 8**
- Support number: **Choose the available Number**
- PSTN Region: **Default**

Phone numbers

This is the phone number customers will call to reach your business. You can select multiple from the numbers you have configured under V

Number	Webex Calling location	Support number
1	RTP Location 8	+19842906061

Add more

- After adding these settings, click "Create" to finalize the channel.
- The way an incoming call is presented to an agent in agent-based routing is via the entry point to the routing flow and the flow mapped to the queue which has the team the agent belongs to.
- With this logic, let's check the queue and make the necessary configuration.
- In the Contact Center navigation pane, under Customer Experience, select Queues.

CUSTOMER EXPERIENCE

Channels

Queues

Business Hours

Audio Files

Flows

AI Agents

Call Recording Schedules

Functions

Surveys

- Click the **Create a Queue** button

Create a queue

- Configure the new queue by providing the name **WebexOne_Queue_[num]** and ensuring these settings match:
- Contact Direction : Inbound Queue
- Channel Type : Telephony

General

Name *

WebexOne_Queue_1

Description

Enter a description

Contact direction *

Inbound queue

You can't change contact direction once the queue is created

Channel type *

Telephony

You can't change channel type once the queue is created

- In Contact Routing Settings, confirm these settings:
- Agent Assignment: Teams
- Routing Pattern: Longest available

Contact routing settings

Skills-based routing

Use skills-based routing for this queue

Turn this option on to route contacts to agents with a required skill set. You can't change this once you've created the queue.

Agent assignment *

Choose how agents are assigned to the queue. You can't change this once you've created the queue.

Teams

Agents

Routing pattern *

The routing pattern determines how to select an agent from other assigned agents in the queue.

Longest available

- For, Call Distribution: In group 1, click on the action and ensure the team **WebexOne_Team_[num]** is added.

Call distribution *

Create a call distribution group to add a team or more to this queue. To distribute a call to more teams, create multiple groups. Be aware that call distribution is independent of other flow-defined queue logic.

Create a group

Group Details

	Priority	Group Name	Switching Time	Actions
⋮	1	Group 1	NA	✎

Group 1

Priority

1

Add group after

0

 Seconds

team

16 Records

Team	Site	Team type
capacity_team	Site-1	CAPACITY
ChrisTeam1	Site-1	AGENT
CiscoLiveTeam	CiscoLiveSite	AGENT
CL_CDS_Team	Site-1	AGENT
D365 Team	Site-1	AGENT

Cancel

Save

- In Advanced Settings, ensure that these values are accurate:
- Service Level Threshold: **30 seconds**
- Maximum Time in Queue: **30 seconds**
- Default Music in Queue: **defaultmusic_on_hold.wav**

Advanced settings

Service monitoring

Allow recording

Recording Enabled at Tenant level

Record all calls

Recording Enabled at Tenant level

Allow pause/resume for calls

Service level threshold

*

60

 Seconds

Maximum time in queue

*

30

 Seconds

Note: Ensure the maximum time in queue setting doesn't conflict with queue logic defined in flows.

Default music in queue

*

defaultmusic_on_hold.wav

Time zone (routing strategies only)

Default (tenant timezone)

- Click Save to finalize the queue.

- It's time to now test the call. To do this, log in to the Agent Desktop using the provided credentials.
- URL: <https://desktop.wxcc-us1.cisco.com/>
- Username: **Contact the lab proctor if information is unavailable.**
- Password: **Contact the lab proctor if information is unavailable.**
- Please select **Desktop** as the telephony option, set the Team as **WebexOne_Team_[num]** , and log in.
- Ensure that the agent is in **Available** Status on the desktop.
- Place a call from your cell phone to the Dialed Number assigned to your entry point and select option 0 to reach the logged-in agent.
- However, you will notice the call is never presented to the agent, and the caller hears music on hold.
- Let's troubleshoot to see why the call is not reaching the agent. The best way will be to look at the flow to see what is happening with the call.
- In the browser tab/window containing Contact Center navigation pane select "Flows" under customer experience section and search for the flow you have mapped to the entry point, **WebexOne_Flow_[num]**

CUSTOMER EXPERIENCE

Channels

Queues

Business Hours

Audio Files

Flows

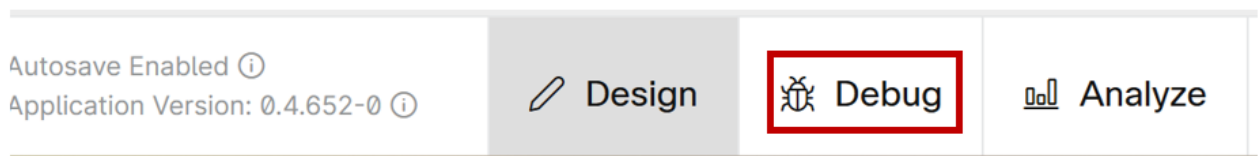
AI Agents

Call Recording Schedules

Functions

Surveys

- Click on the "Debug" option and review the last call.



- You will notice that the call passed through the "Menu" node, where option "0" was pressed.

Interactions			4f603c0c-08b0-42b5-a0d0-0d72540be408 ×
Sequence	Activity Name	Outcome	
1	NewPhoneContact	Success	<pre>[[{"type": "ss", "value": "Welcome to the Webex One Contact Center team. Internal transfer to a different team.\n", "name": "Welcome to the Webex agent, and 1 to perform an internal transfer to a different team.\n"}]]</pre> Make Prompt Interruptible : true Connector : Cisco Cloud Text-to-Speech Speaking Rate : 1 Override Default Language & Voice Settings : Output Voice : Enable decryption : false
2	Menu_ja4	Success	
3	QueueContact_hqu	Success	
4	PlayMusic_xze	Success	
5	⚡ ContactEnded	Success	
			Activity Outputs OptionEntered : 0
			Modified Variables

- Then, the call landed in the queue node where the selected queue had the ID "4d0282c7-6478-444b-b41a-8e331614acae".

Interactions			4f603c0c-08b0-42b5-a0d0-0d72540be408 ×
Sequence	Activity Name	Outcome	
1	NewPhoneContact	Success	Contact Priority Variable : Queue Variable : priorityRadioGroup : skills : Enable decryption : false Queue : 4d0282c7-6478-444b-b41a-8e331614acae agentAvailabilityRadioGroup : staticAgentAvailability
2	Menu_ja4	Success	
3	QueueContact_hqu	Success	
4	PlayMusic_xze	Success	
5	⚡ ContactEnded	Success	
			Activity Outputs QueueId : 4d0282c7-6478-444b-b41a-8e331614acae FailureCode : FailureDescription :
			Modified Variables

- Now, the call moves to the "Play Music" node, where "defaultmusic_on_hold.wav" is played instead of being presented to the logged-in agent.

Interactions			4f603c0c-08b0-42b5-a0d0-0d72540be408 ×
Sequence	Activity Name	Outcome	
1	NewPhoneContact	Success	Activity Name : PlayMusic_xze
2	Menu_ja4	Success	
3	QueueContact_hqu	Success	
4	PlayMusic_xze	Success	
5	⚡ ContactEnded	Success	
			Activity Inputs Music Duration : Music File : defaultmusic_on_hold.wav Start Offset : 0 Dynamic Audio File : audioRadioGroup : staticAudio Enable decryption : false
			Activity Outputs
			Modified Variables

- To figure out what queue it is in the Control Hub, go to the "Queues" section and in the URL, append the queue ID "/4d0282c7-6478-444b-b41a-8e331614acae" noted before and load the page.

admin.webex.com/wxcc/customer-experience/queues/4d0282c7-6478-444b-b41a-8e331614acae

webex Control Hub

Configure your organization's recording retention policy in Control Hub to enable purging of recordings and transcripts. Start

< Main Menu

Contact Center

Overview

CUSTOMER EXPERIENCE

Channels

Queues

Business Hours

< Queues

WebexOne_Queue_Anuj

ID: 4d0282c7-6478-444b-b41a-8e331614acae • Last Modified: September 22, 2025 16:35 PM

General

Name * WebexOne_Queue_Anuj

Description Enter a description

- The queue page that loads is "**WebexOne_Queue_Anuj**" which is not the correct queue for the agent who is logged in.

< Queues

WebexOne_Queue_Anuj

ID: 4d0282c7-6478-444b-b41a-8e331614acae • Last Modified: September 22, 2025 16:35 PM

General

Name * WebexOne_Queue_Anuj

Description Enter a description

Contact direction * Inbound queue

You can't change contact direction once the queue is created

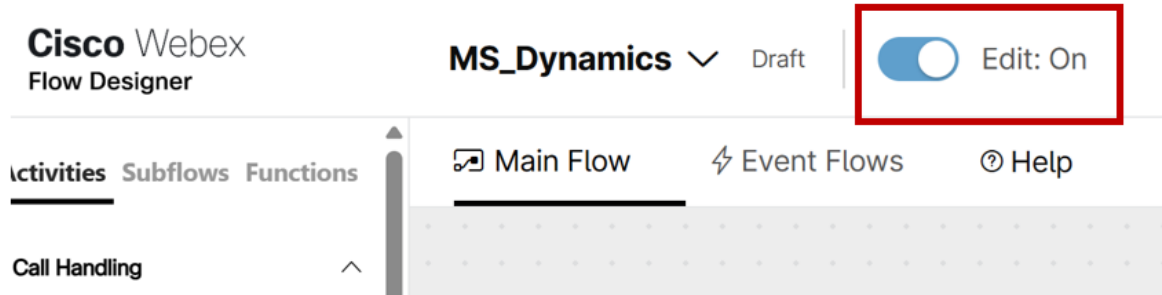
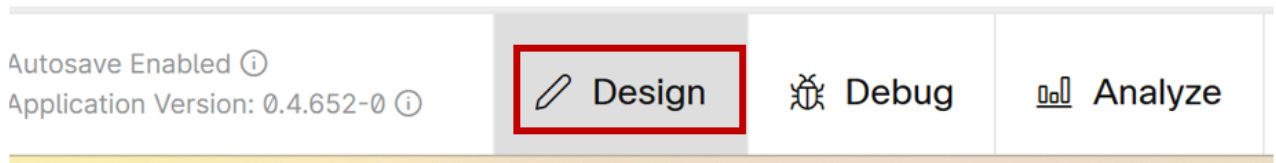
Channel type * Telephony

You can't change channel type once the queue is created

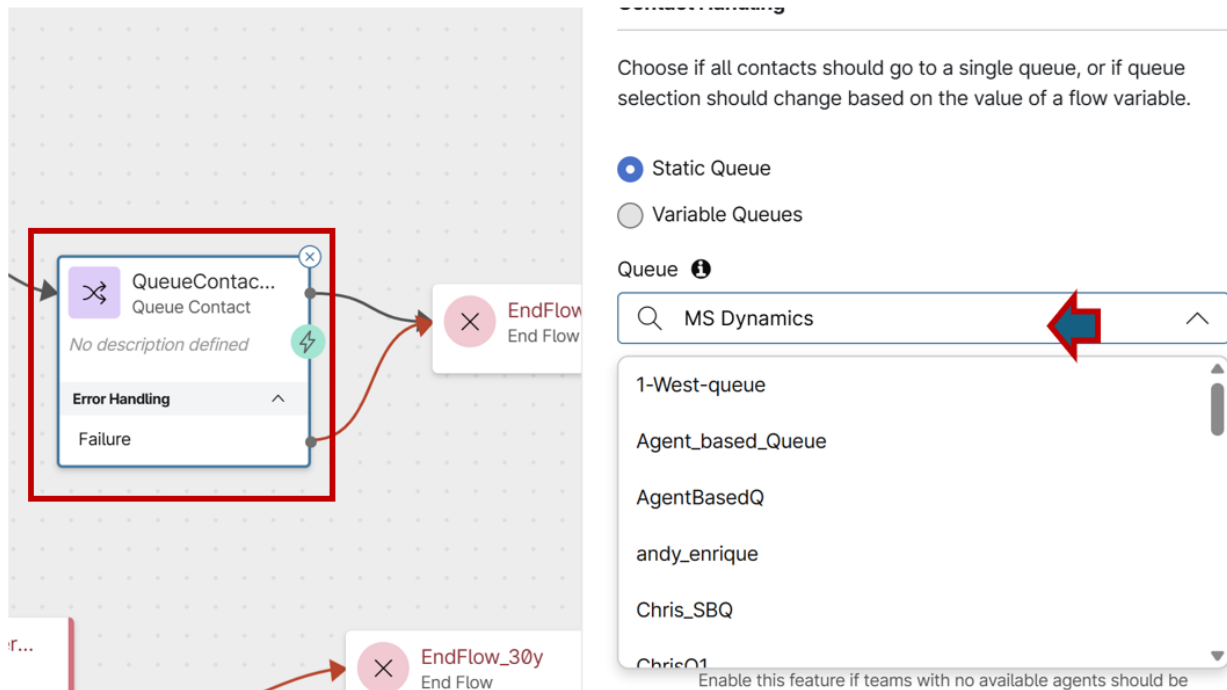
Referenced by Go to the list to view all references by entity type.

Reference list

- To correct this, come back into the Design section of the flow and click on the "Edit" option in the flow.



- Select the "Queue contact" node, and from the drop-down menu, select the queue that has your team and the agent i.e **WebexOne_Queue_num]**



- Toggle "Validation" to "On" to ensure there are no validation errors, and then publish the flow.

Validation Details

Flow Errors:0

Flow has no errors and can be published

Recommendations: 2

1. Add descriptions for activities
2. Ensure that all flow paths terminate. Event Flows are not required, however any flow paths connected to Event Handlers must terminate. Use multiple terminating activities as needed to keep the canvas organized.

MS Dynamics

Select a single queue used for all interactions in this branch of the flow. If the queue supports Skills Based Routing, additional configurations appear below.

☐

Set Contact Priority

Enable this feature if contacts in this path of the flow should be prioritized over other contacts in the queue. If this is enabled for some but not all contacts, there is a risk that those without an assigned priority will never move to the top of the queue.

☐

Check Agent Availability

Enable this feature if teams with no available agents should be excluded from routing as time in queue progresses. A Queue's Call Distribution Group(s) may be skipped to find an agent sooner.

☒ Flow Errors: 0

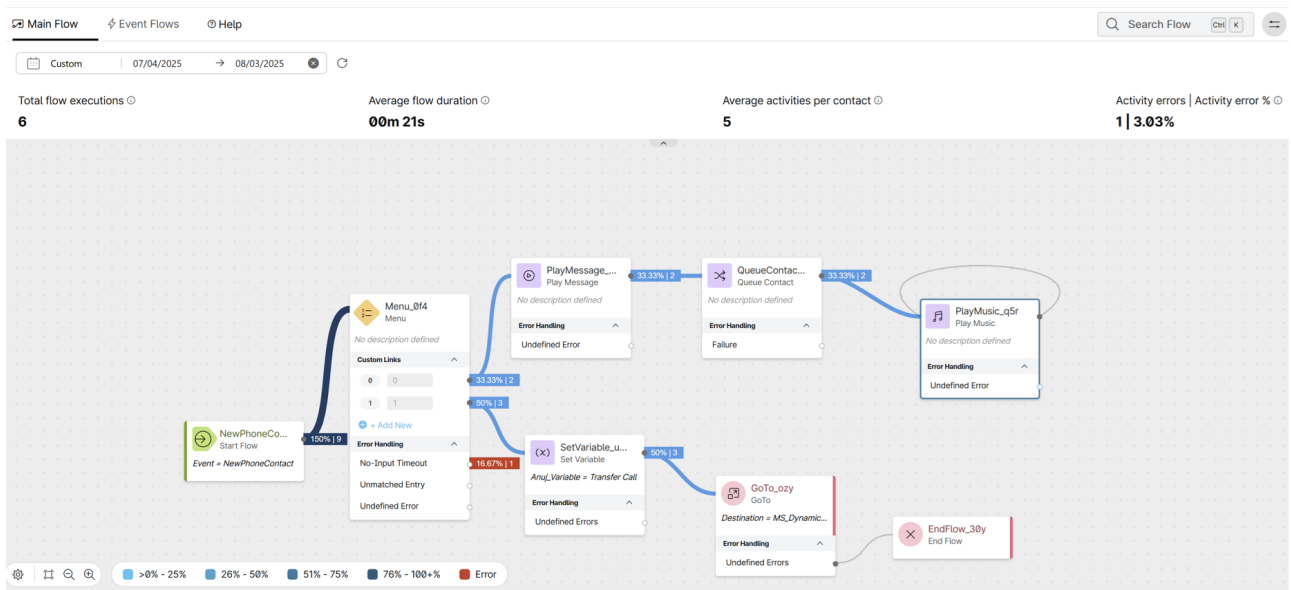
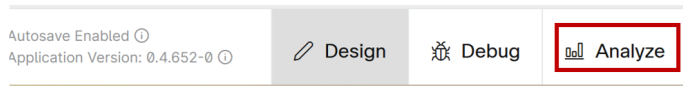
☒ Autosave

☒ Validation: On **Publish Flow**

Note

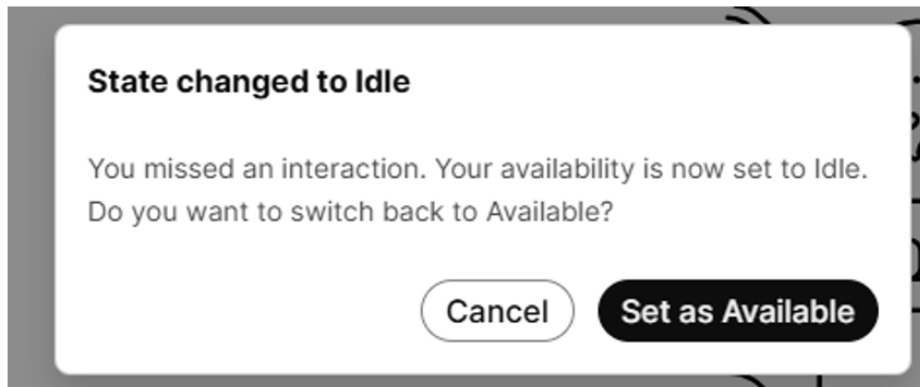
When publishing, it is important to ensure the label you have set is the same one that was set when the entry point was being mapped to the routing flow.

- Place another call from your cell phone to the dialed number you have assigned and select option 0 to successfully reach the logged-in agent.
- In the Flow Canvas you can now also check how the calls have traversed through various nodes by clicking on the Analyze capability and selecting the current day.

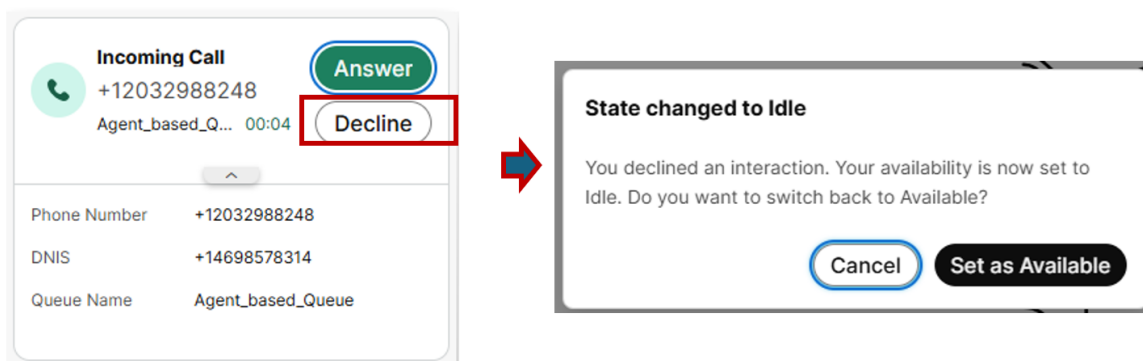


1.2.4 Section 3: Analyze Agent Performance with Call Reports

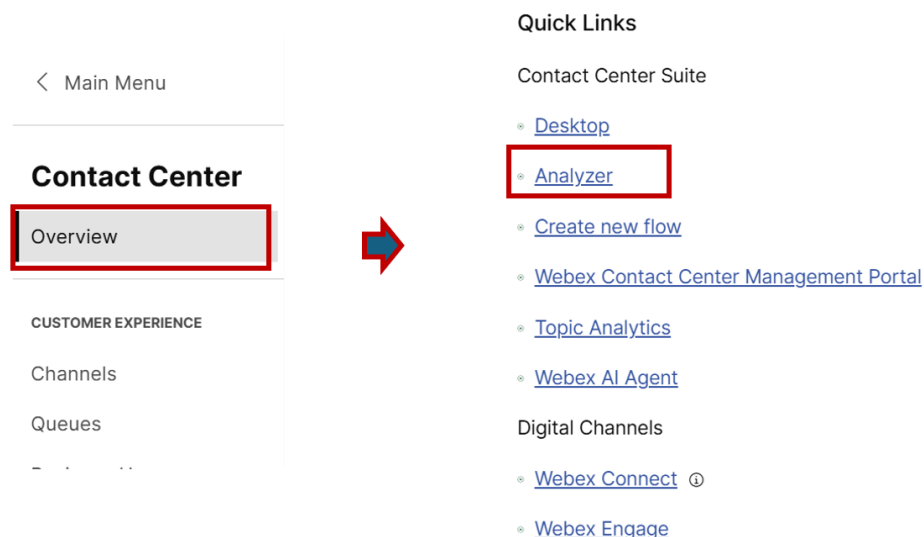
- The goal of this lab section is to review reports to determine what happened with few calls.
- To create a dataset of calls, let's simulate some common agent-side issues.
- First, ensure that the agent is ready and make a call from your cell phone.
- When the call is presented to the agent, do not accept it. Let the call ring for 12 seconds until it goes to a "no answer" state.
- Hangup to disconnect the call.



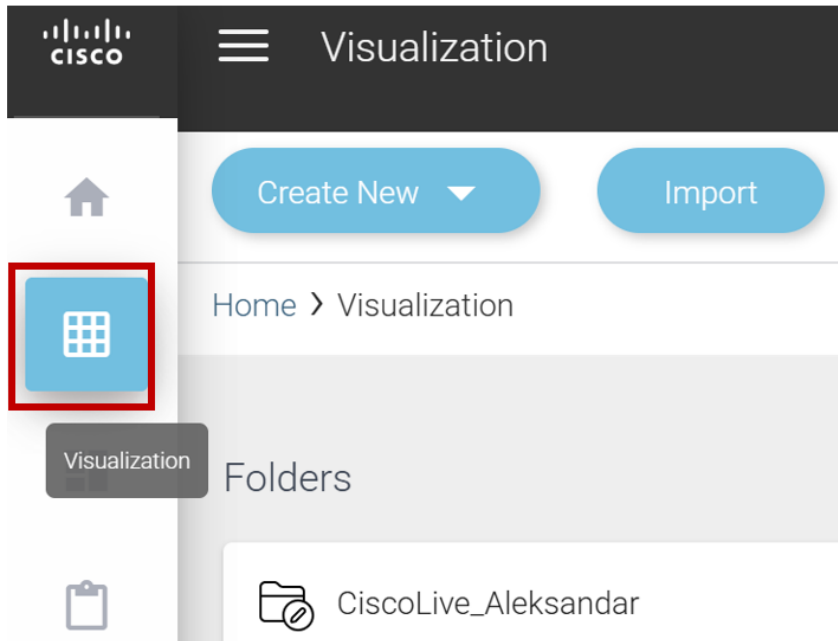
- Next, make another call.
- This time, when the call is presented to the agent, reject the call. Disconnect call from your cell phone.



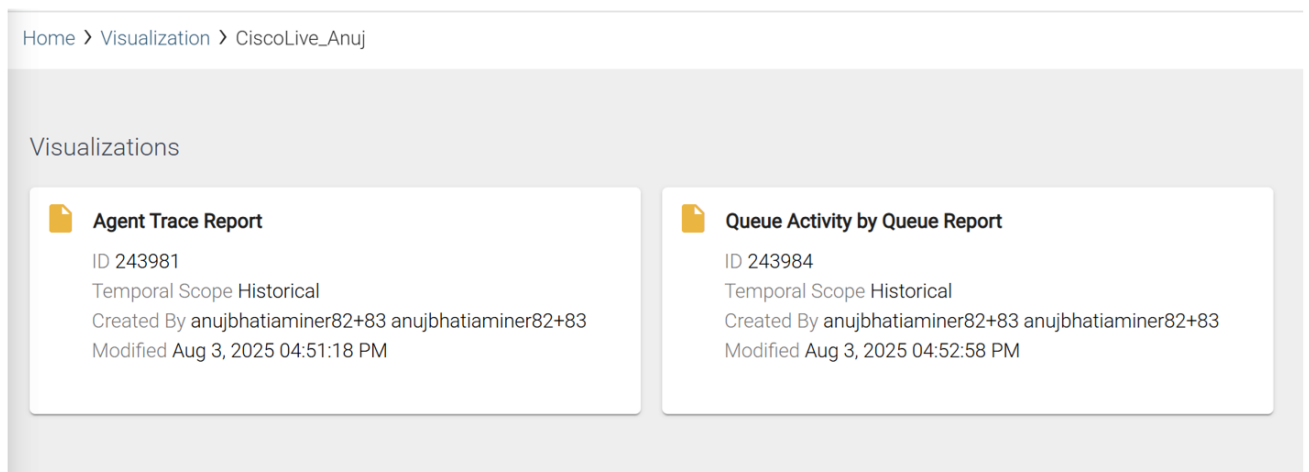
- Now that you have a variety of calls in the system, you can explore the reports to see how to track "**Contact Offer**" and "**Assignment Failures**" to the agent.
- In Control Hub under "Contact Center" select "Overview." In the "Quick Links" section on the right, select "Analyzer."



- Click on "Visualization" and double-click the folder **WebexOne_Report_User[num]** containing your user details.



- There are two reports in the folder which are copy of the stock report:
- **"Agent Trace Report"** and
- **"Queue Activity By Queue Report"**.



- Each report includes three columns:
- **RONA Count:** This captures all calls that failed because the configured RONA timer expired in WxCC (reasonCode: RONA_TIMER_EXPIRED) or Because of a configured ring timeout on the device (reasonCode: NO_ANSWER_USER).
- **Call Reject Count:** This tracks all calls that failed because the agent explicitly declined the call (reasonCode: USER_DECLINED).
- **Offer Error Counts:** This reflects call offer and assignment failures to the agent caused by any other error.
- To review the "Agent Trace" report, first click on the "Edit" option

Agent Trace Report

ID 243981
Temporal Scope Historical
Created By anujbhatiaminer82+83 anujbhatiaminer82+83
Modified Aug 3, 2025 04:51:18 PM

Queue Activity

- Run
- Edit**
- Details
- Export as Excel
- Export as CSV
- Export Template

- Ensure that the start time of the report is set for "Today."

Modules

Formatting

Module

+

Start Time

Today

If run today:

Start Date: **2025-08-03**

End Date: **2025-08-03**

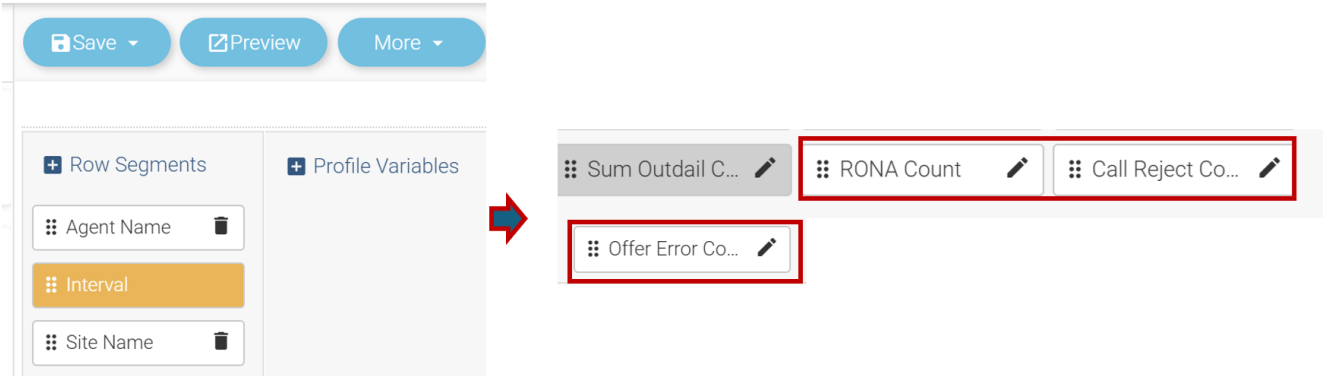
Including

All Days

Start Day of the Week ⓘ

Monday

- In the profile variables, you can also confirm that the three variables i.e. **RONA Count**, **Call Reject Count** , and **Offer Error Counts**—are present.



- Save the visualization and Click the "Preview" option.
- You should now see counts for "RONA" and "Call Reject." Click on the value and the magnification symbol for a further drill-down of the number.

Cisco Analyzer Agent Trace Report						
Agent Name: All		Site Name: All		Team Name: All		Time Zone: (UTC-04:00) Eastern Daylight Time (America/New_York)
Interval: Daily		Duration: Today				
Blind Transfer	Inbound Average Handle Time	Outdial Average Handle Time	RONA Count	Call Reject Count	Offer Error Count	
0.0	00:00:00	00:00:00	0	0	0	
0.0	00:00:00	00:00:00	0	0	0	
0.0	00:00:00	00:00:00	0	0	0	
0.0	00:00:00	00:00:00	0	0	1	
0.0	00:00:00	00:00:00	0	0	0	
0.0	00:00:00	00:00:00	0	0	0	
0.0	00:00:00	00:00:00	0	0	0	
0.0	00:00:00	00:00:00	1	1	0	

- To easily review the calls in the report, use the Custom Select option in Agent Name to search for the designated user and their calls

Agent Name - Custom Selection

Agent Name picked

Selected : 0

Agent Name List

☐ Select All

☐ N/A

- A new window will pop up with all the details of this call. As needed, you can add fields and measures from the available options to get all the desired information related to the call and the agent.

Drill Down Call Reject Count

Search

Fields

Activity State
Agent Endpoint (DN)
Agent ID
Agent Login
Agent Name
Agent Session ID
Agent System ID
Call Back Type
Changed By ID
Changed By Name
Channel ID
Channel Type

Measures

Interaction ID	Call Direction	Queue Name	Reason Code	Event Name
8ea6cfe-d43a-4708-b147-8...	Inbound	Agent_based_Queue	USER_DECLINED	con-to-agent-error

1 to 1 of 1

Page 1 of 1

- To test this, you can click on "Agent ID" and it will be appended to the end of the report as a new column.
- The same process can be used to explore the "Queue Activity By Queue" report.

Congratulations!! on completing this section of the lab!

You've successfully navigated the complexities of an inbound call flow configuration, debugged common issues, and learned how to analyze agent performance by simulating and tracking different call outcomes in reports.

1.3 Lab 2 - Seamless Call Transfer & Preserving flow Variables

Please use the following credentials to connect to Control Hub and configure Webex Contact Center:

Control Hub URL	https://admin.webex.com
Username	labuser ID @wx1.wbx.ai (where ID is your assigned pod number; this ID will be provided by your proctor)
Password	webexONE1!

1.3.1 Objective

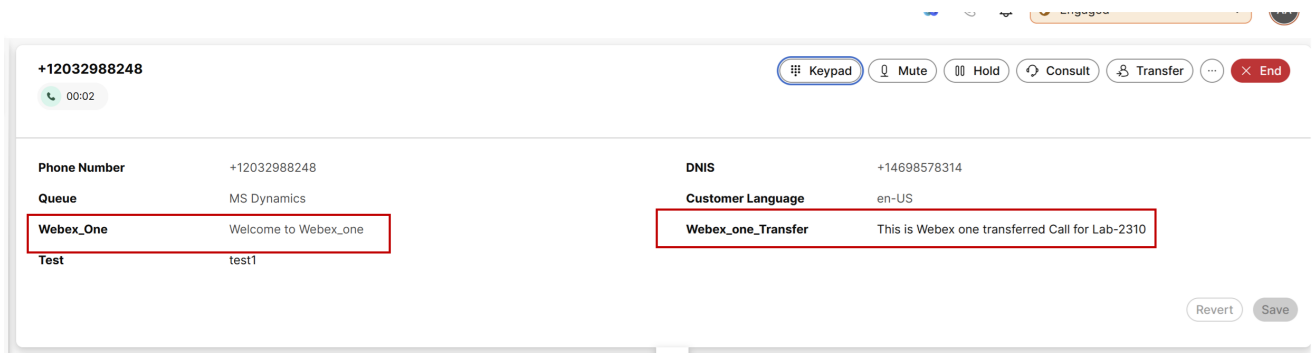
The objective of this exercise is to correct a call flow that currently loses global variables during a transfer.

The current flow is as follows:

- A customer calls and presses option 1, which transfers the call to another department. This transfer causes the call to leave the system and re-enter through a new entry point, creating a new call leg. Within this new flow, the customer presses the user ID option to reach the correct team and agent.
- This process presents two main issues for the business:
- The call is unnecessarily leaving and re-entering the system.
- All original global variables are lost during this transfer, which prevents the agent from seeing crucial information.
- The task is to correct this flow so that the call is transferred seamlessly to the correct destination while preserving all original global variables for the agent.

1.3.2 Section 1 : Experience the Issue

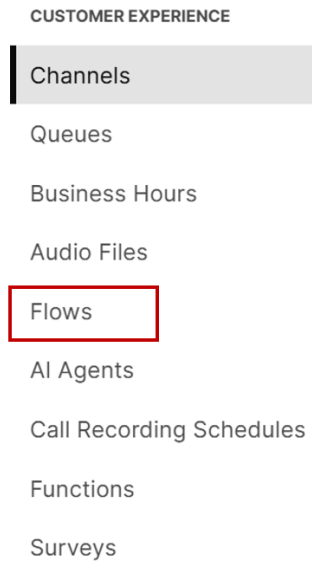
- Ensure that the agent is logged into the desktop and has an "Available" status.
- Call the same entry point number that was configured in Lab 1 for an inbound call and press option 1.
- You will hear a **ringback** and a menu prompting you to enter your user ID.
- The **ringback** you hear is the **first issue**, as it indicates a new call is being made and hitting a transferred entry point (DN).
- Once you enter your user ID, the call will be presented to the agent.
- Accept the call.
- On the desktop, you should see two flow variables, "**Wbex_one_Tansfer**" and "**Webex_one**" However, you will notice that they are not appearing.



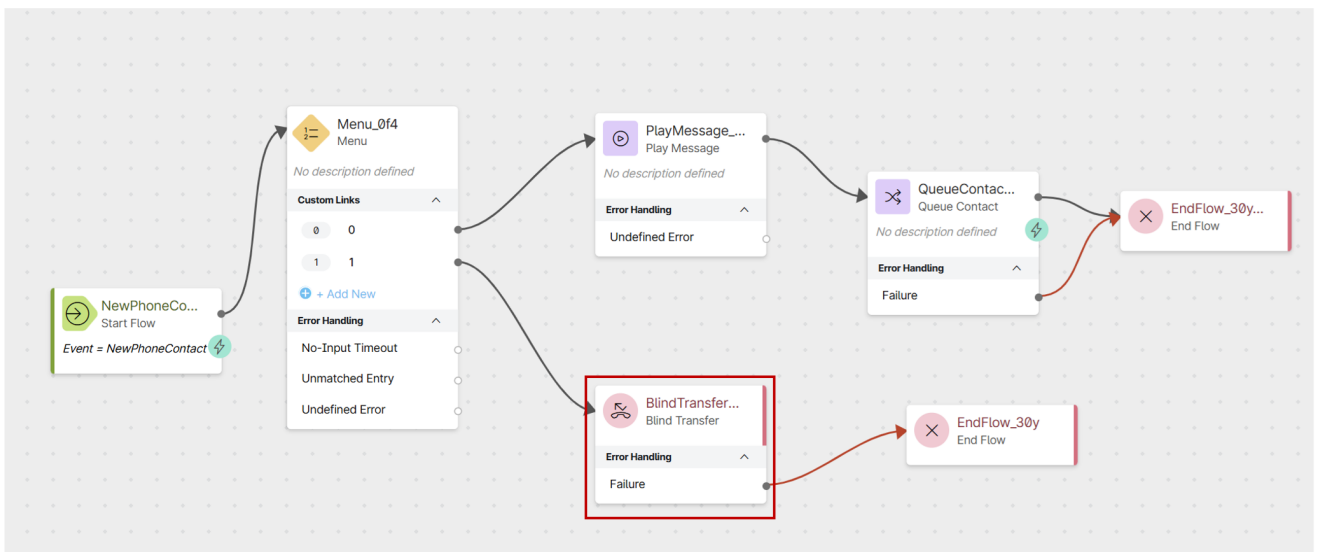
- **This is the second problem:** The original flow variables collected in the first flow are being dropped during the transfer.

1.3.3 Section 2 : Inspect the Flow

- To understand why this is happening, let's inspect the first flow.
- In the Customer Experience section of Control Hub, select "Flows" and search for the flow you have mapped to your entry point, "**WebexOne_Flow_[num]**."



- In the menu node for Option 1, you will notice that it maps to a "**Blind Transfer**" node.



- Select the Blind Transfer node, and you will see it is transferring the call to a number "**19842906070**", which is mapped to a different flow called "**WebexOne_Flow_Transfer**"

Transfer Dial Number ^

Indicate the Dial Number (DN) that the call is transferred to. This can be a specific number that is manually entered, or a dynamic number that is indicated through a flow variable.

☒ Specific Dial Number

19842906070

☐ Variable Dial Number

- This is the core of the problem. We are instructing the flow to transfer the call to an external number, which creates a new call leg.
- As a result, the variables as well defined in this flow are not being passed to the "**WebexOne_Flow_Transfer**" flow.
- To check these variables, click on any empty space on the flow canvas and review the right side of the flow in **Variable Definition** section.

Variable Definition ^

Configuration

Desktop Viewability & Order

Custom Variables

Flow Variables are fully configurable local variables. They can never be viewed in reports, but can optionally be agent viewable.

Flow Variables ⌵

View All Variables ⌵

🔊 Webex_one_Transfer ×

🔊 Webex_One ×

Add Flow Variable

1.3.4 Section 3 : Correct the Flow

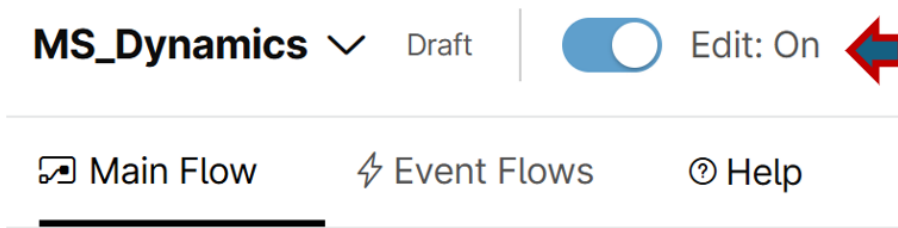
- To rectify this, we need a node that handles internal transfers more effectively.

- The Flow Designer provides a "GoTo Node" specifically for this use case.

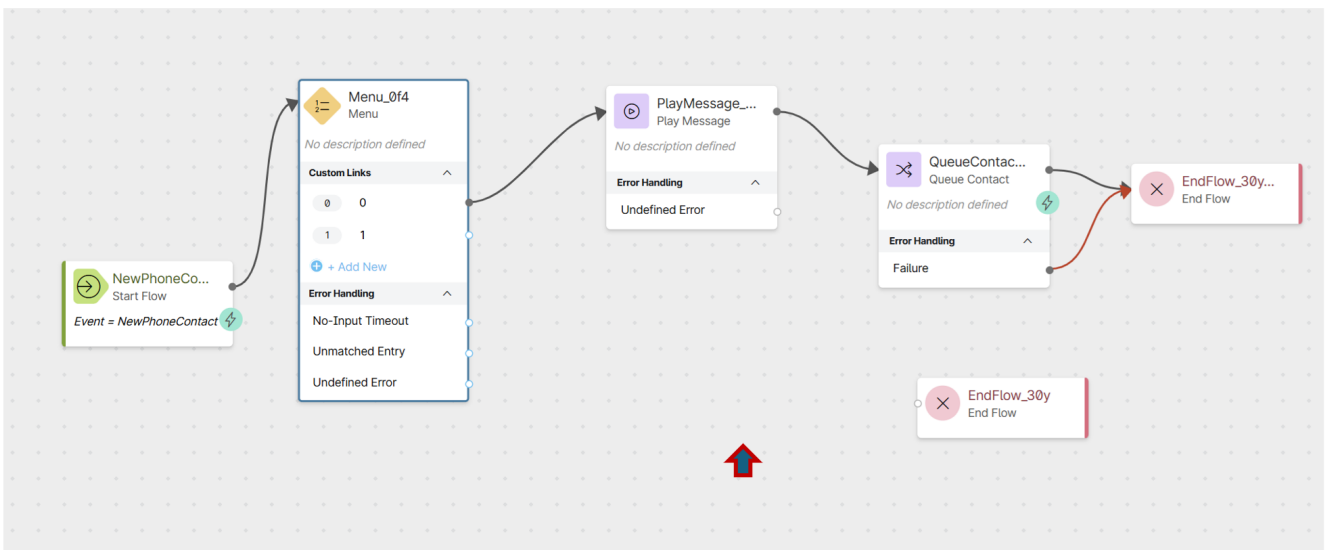
 Note

A GoTo Node is used to seamlessly transfer a call to another flow within the same system, preserving variables. A Blind Transfer node is used to transfer a call to an external number.

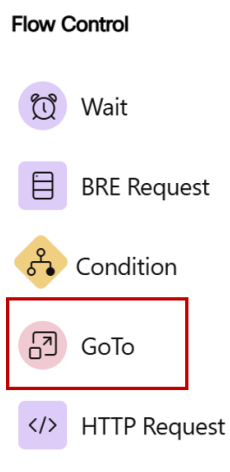
- To replace the Blind Transfer node, first enable the edit mode of the flow designer.



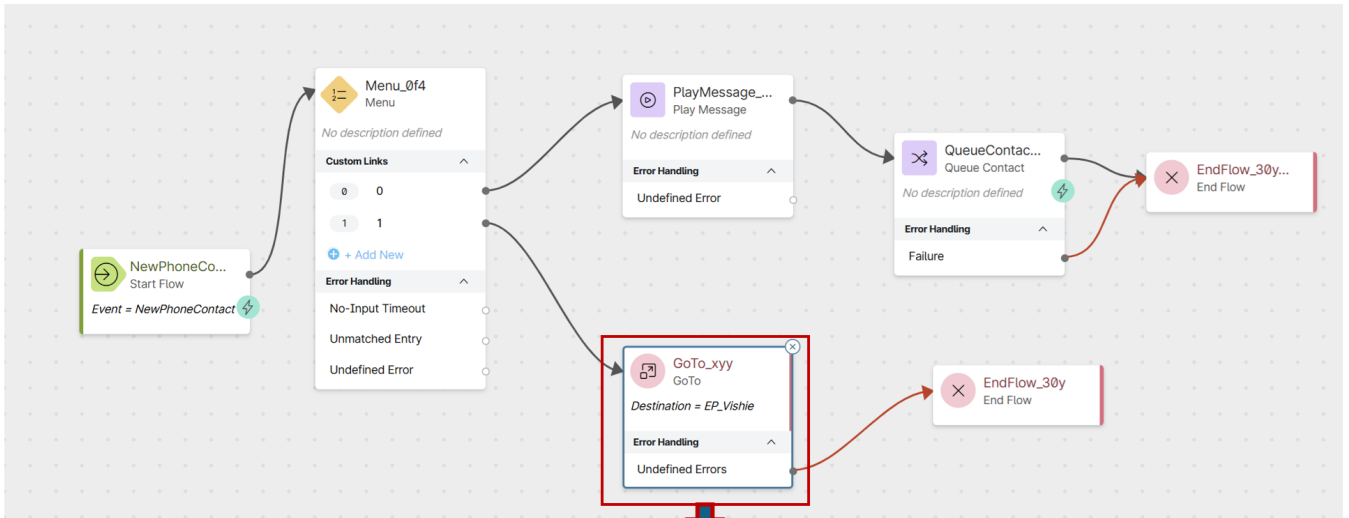
- Delete the Blind Transfer node by selecting it and pressing the delete key.



- From the Flow Control section, drag a "GoTo Node" onto the canvas.



- In the GoTo Node's properties, select "Entry point" as the destination type and map it to "**WebexOne_Transfer_EP**"



Flow Destination Settings

Indicate if the current flow should go to an entry point or to another flow. If going to another flow, a Variable Mapping section appears for additional configuration.

Destination Type

Entry Point

☒ Static Entry Point ⓘ

☐ Dynamic Entry Point ⓘ

Entry Point



Webexone_transfer_EP

Select an entry point if the flow logic should change based on the active

- Connect the Menu node's Option 1 output to the GoTo Node and the "undefined" error output to the "End of the flow" node, as shown in the screenshot above
- Toggle "Validation" to "On" to ensure there are no validation errors, and then publish the flow.

Validation Details

Flow Errors: 0

Flow has no errors and can be published

Recommendations: 2

1. Add descriptions for activities
2. Ensure that all flow paths terminate. Event Flows are not required, however any flow paths connected to Event Handlers must terminate. Use multiple terminating activities as needed to keep the canvas organized.

MS Dynamics

Select a single queue used for all interactions in this branch of the flow. If the queue supports Skills Based Routing, additional configurations appear below.

☐ Set Contact Priority

Enable this feature if contacts in this path of the flow should be prioritized over other contacts in the queue. If this is enabled for some but not all contacts, there is a risk that those without an assigned priority will never move to the top of the queue.

☐ Check Agent Availability

Enable this feature if teams with no available agents should be excluded from routing as time in queue progresses. A Queue's Call Distribution Group(s) may be skipped to find an agent sooner.

Flow Errors: 0

Autosave

Validation: On

Publish Flow

1.3.5 Section 4 : Verify the Solution

- Ensure that the agent is logged into the desktop and has an "Available" status.
- Call your provided number from your cell phone again and press Option 1.
- You should hear no ringback, but instead, be directly presented with the menu option to enter your user ID.
- This resolves the first problem of creating a new call leg for the transfer.
- Once you provide the user ID, the agent should receive the call.
- After the call is accepted, you should now see the two variables defined in the first flow on the agent desktop.

The screenshot shows a call center agent's desktop interface. At the top, there's a header bar with a phone icon, the number +12032988248, and a timer showing 00:02. Below this is a row of buttons: Keypad, Mute, Hold, Consult, Transfer, and End. The main area displays call details in a table-like format. The variables Webex_One and Webex_one_Transfer are highlighted with red boxes. At the bottom right, there are Revert and Save buttons.

Phone Number	+12032988248	DNIS	+14698578314
Queue	MS Dynamics	Customer Language	en-US
Webex_One	Welcome to Webex_one	Webex_one_Transfer	This is Webex one transferred Call for Lab-2310
Test	test1		

- This confirms that the second problem of variables being lost has also been resolved.

Congratulations !! on successfully completing this exercise!

You've learned how to perform seamless internal transfers within entry points and now understand the crucial difference between a Blind Transfer and a GoTo Node.

1.4 Lab 3 - Set Up Outdial Calls in WxCC

Please use the following credentials to connect to Control Hub and configure Webex Contact Center:

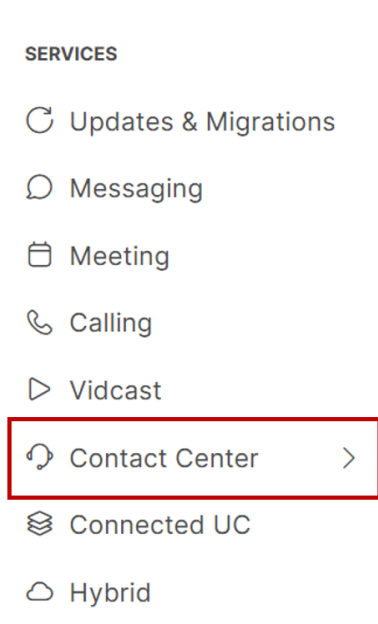
Control Hub URL	https://admin.webex.com
Username	labuser ID @wx1.wbx.ai (where ID is your assigned pod number; this ID will be provided by your proctor)
Password	webexONE1!

1.4.1 Objective

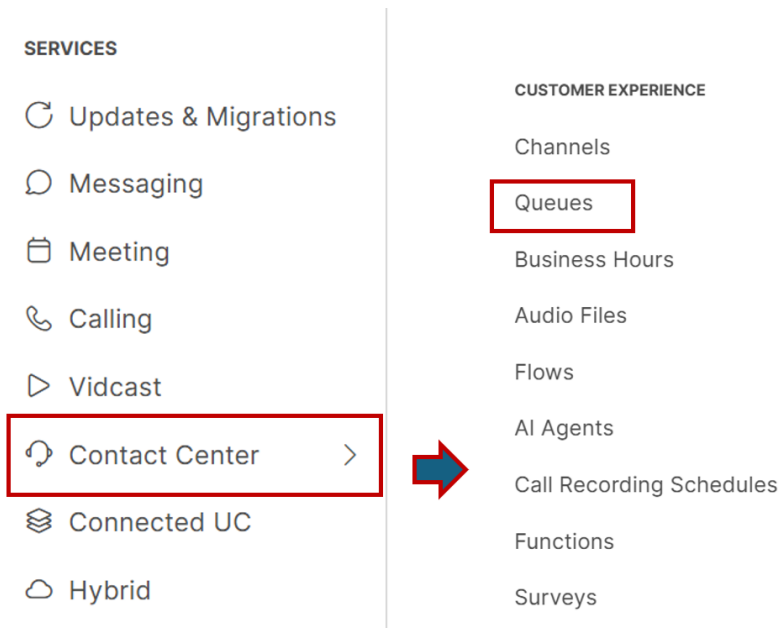
In this lab exercise, the audience will gain hands-on experience configuring WxCC solutions to enable outdial capabilities from scratch. Furthermore, the exercise will also equip participants with essential debugging skills to identify and rectify frequent implementation errors ensuring a robust and functional deployment.

1.4.2 Section 1 : Setup Outdial

- Log into Webex Control Hub with the provided credentials.
- In Control Hub, navigate to Services and click on Contact Center.



- In the Contact Center navigation pane, under Customer Experience, select **Queues**.



- Create a new queue by clicking on the "Create a Queue" option.
- The Queue Creation Wizard will appear. Provide the following details:
- **General**
- **Name:** [Provide a descriptive name for your queue]
- **Contact direction:** Outdial Queue
- **Channel type:** Telephony

The screenshot shows the 'General' tab of the Queue Creation Wizard. The 'Name' field is highlighted with a red box. The 'Description' field is empty. The 'Contact direction' is set to 'Outdial queue' and the 'Channel type' is set to 'Telephony'. Below these fields, there are two messages: 'You can't change contact direction once the queue is created' and 'You can't change channel type once the queue is created'.

- **Contact Routing Settings**
- **AgentAssignment:** Teams
- **Routing Pattern:** Longest available
- **Call Distribution:** Create a group and add the team **WebeOne_Team_[num]**

Contact routing settings

Outbound campaign ⓘ ☒ Outbound Campaign is only applicable for Outdial Queues.

Skills-based routing ☒ Use skills-based routing for this queue
Turn this option on to route contacts to agents with a required skill set. You can't change this once you've created the queue.

Agent assignment *
Choose how agents are assigned to the queue. You can't change this once you've created the queue.

Teams ⓘ

Agents ⓘ

Routing pattern *
The routing pattern determines how to select an agent from other assigned agents in the queue.

Longest available ⓘ

Call distribution *
Create a call distribution group to add a team or more to this queue. To distribute a call to more teams, create multiple groups. Be aware that call distribution is independent of other flow-defined queue logic.

Create a group

Group 1

Priority

Add group after ⓘ Seconds

2 Records

Team ↑	Site	Team type
☐ MS Dynamic	Site-1	AGENT
☐ MS Dynamics2	Site-1	AGENT

• Advanced Settings

- **Service level threshold:** 30 seconds
- **Maximum time in queue:** 30 seconds
- **Default music in queue:** defaultmusic_on_hold.wav

Advanced settings

Service monitoring ☒

Allow recording ☒
Recording Enabled at Tenant level

Record all calls ☒
Recording Enabled at Tenant level

Allow pause/resume for calls ☒

Service level threshold ⓘ * Seconds

Maximum time in queue * Seconds

Default music in queue *

Time zone (routing strategies only)

- Once these settings are added, click Create to finalize the queue.
- Lets create a Entry point to map the queue to this entry point.
- Navigate back to Customer Experience in Contact Center and click on **Channels**.

CUSTOMER EXPERIENCE

Channels

Queues

Business Hours

Audio Files

Flows

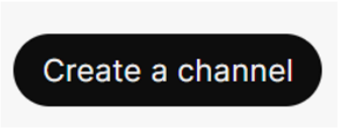
AI Agents

Call Recording Schedules

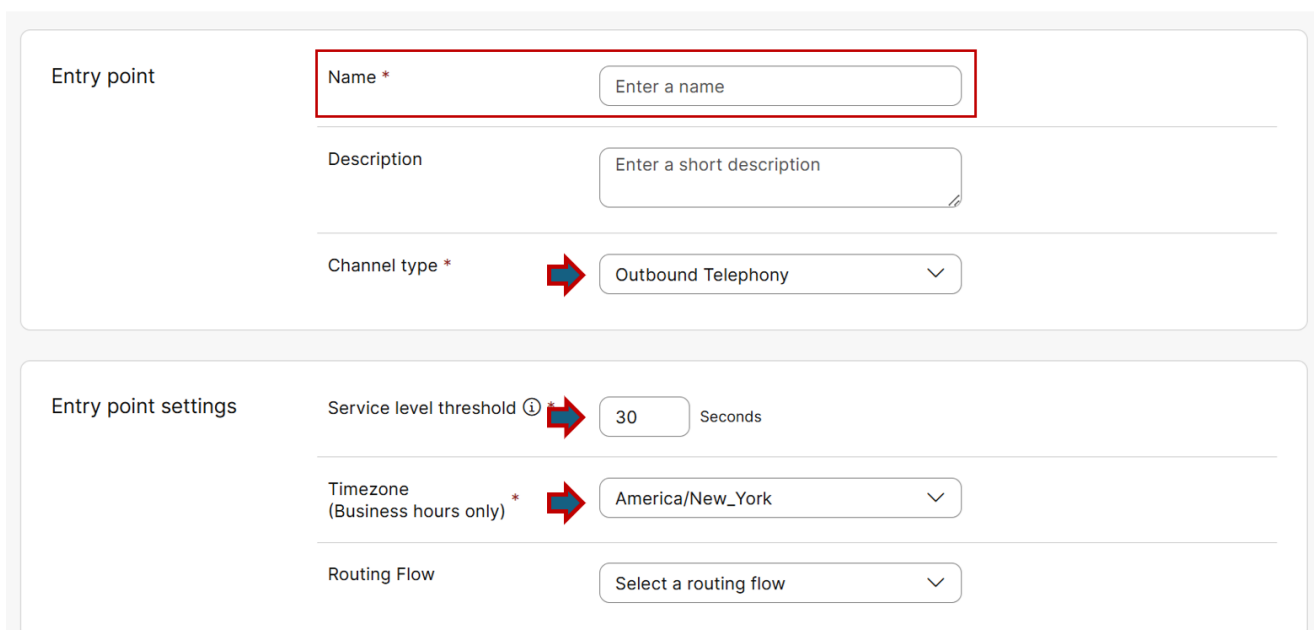
Functions

Surveys

- Create a new channel by clicking on the "Create a channel" option.



- The Channel Creation Wizard will appear. Provide the following details:
- **Name:** [Provide a descriptive name for your channel]
- **Channel type:** Outbound Telephony
- **Service level threshold:** 30 seconds
- **Timezone:** America/New York



The form is divided into two main sections: "Entry point" and "Entry point settings".

Entry point section:

- Name ***: A text input field with the placeholder "Enter a name".
- Description**: A text input field with the placeholder "Enter a short description".
- Channel type ***: A dropdown menu with "Outbound Telephony" selected. A red arrow points to this field.

Entry point settings section:

- Service level threshold**: A field with a value of "30" and the unit "Seconds". A red arrow points to this field.
- Timezone (Business hours only) ***: A dropdown menu with "America/New_York" selected. A red arrow points to this field.
- Routing Flow**: A dropdown menu with the placeholder "Select a routing flow".

- After these settings are added, click Create to finalize the channel.
- Since outdial is an agent activity, the Agent Desktop should have the capability to call any number outside the WxCC ecosystem.
- To enable this capability, create an agent profile and map it to the agent.

- Navigate to Desktop Experience in Contact Center and click on **Desktop Profiles**.

DESKTOP EXPERIENCE

AI Assistant

Multimedia Profiles

Outdial ANI

Desktop Layouts

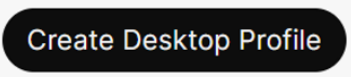
Dial Plans

Address Books

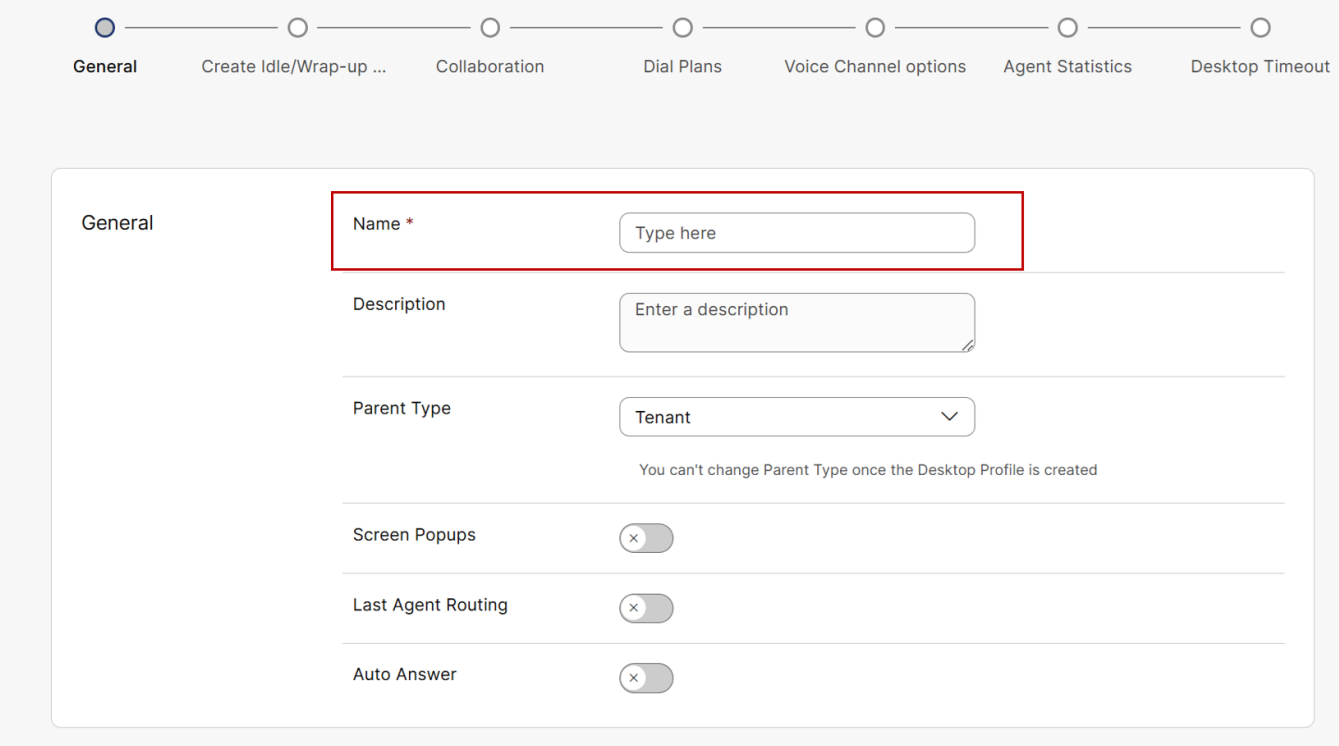
Desktop Profiles

Idle/Wrap-up Codes

- In Desktop Profiles, create a new profile by clicking on "Create Desktop Profile".



- In the General section, provide the desired name for your profile.



General Create Idle/Wrap-up ... Collaboration Dial Plans Voice Channel options Agent Statistics Desktop Timeout

General

Name *

Description

Parent Type Tenant ▼

You can't change Parent Type once the Desktop Profile is created

Screen Popups ☐

Last Agent Routing ☐

Auto Answer ☐

- Move to "Dial Plans" by clicking Next button (at the bottom of the screen) a couple of times.
- Enable "**Outdial**".
- Select the newly create entrypoint as an "**Outdial Entry Point**".
- Select the preconfigured address book "**WebexOne_outdial_AddressBook**" as an "**Address Book**".

General Idle/Wrap-up Codes Collaboration **Dial Plans** Voice Channel options Agent Statistics

Dial Plans

Outdial ☒

Outdial Entry Point * Anuj_Outdial_EP

Address Book * Adressbook1

Outdial ANI None

Dial Plan ☐

- Move to "Voice Channel Options" by clicking Next and ensure that "Desktop" is enabled under "Voice Channels options".

General Idle/Wrap-up Codes Collaboration Dial Plans **Voice Channel options**

Voice Channel options

One Voice option must always be checked.

☒ Agent DN

☒ Extension

☒ Desktop

Validation for Agent DN

☒ Unrestricted (Allow any value)

☐ Provisioned DN (Restrict login DN to provisioned agent DN)

☐ Validate using Dial Plans (Select from list)

- Proceed to the end of desktop profile creation by clicking Next and finally Create.
- Now, Navigate to the User Management section in Contact Center and click on **Contact Center Users**.
- Bring up your user and assign the newly created desktop profile under "Desktop Profile" and Save changes.

USER MANAGEMENT

Sites

Skill Definitions

Skill Profiles

Teams

User Profiles

Contact Center Users

anujbhatiaminer82+83 anujbhatiaminer82+83

ID: 248eb10d-f0d6-45ec-9336-ea7e04066a9a • Last Modified: July 24, 2025 08:29 AM

General

First name

anujbhatiaminer82+83

Last name

anujbhatiaminer82+83

Email

anujbhatiaminer82+83@gmail.com

User Profile *

Administrator Profile

Contact Center *

☒

Referenced by

Go to the list to view all references by entity type.

Reference list

Supervisor Settings

Primary team ⓘ

MS Dynamic

Agent Settings

Site

Site-1

Teams

MS Dynamic

1 Teams

Clear All

Desktop Profile

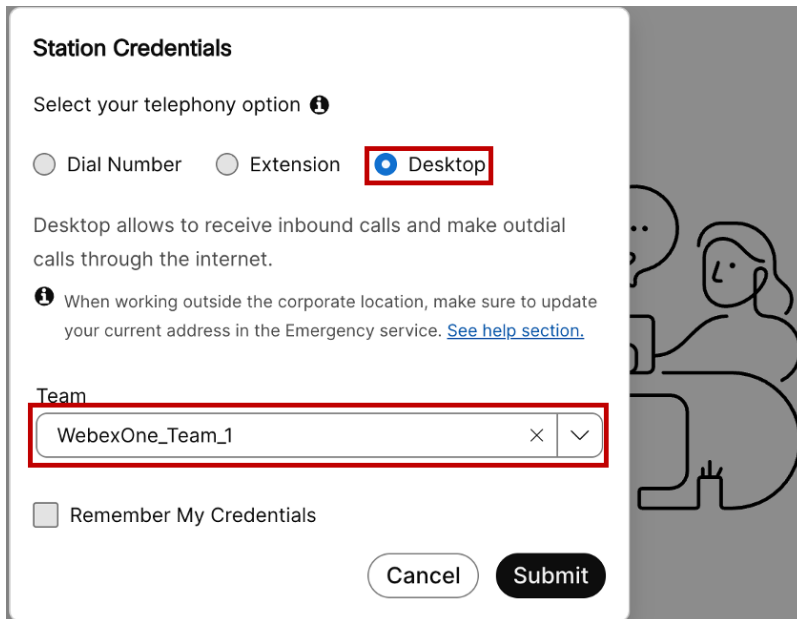
Agent-Profile

Multimedia Profile

Default_Telephony_Profile

1.4.3 Section 2 : Test Outdial

- Now, log in to the Agent Desktop using the provided credentials.
- **URL:** <https://desktop.wxcc-us1.cisco.com/>
- **Username:** Contact the lab proctor if information is unavailable.
- **Password:** Contact the lab proctor if information is unavailable.
- Please select desktop as telephony option and set the Team as **WebexOne_Team_[num]** and login.



Station Credentials

Select your telephony option ⓘ

☐ Dial Number
 ☐ Extension
 ☒ Desktop

Desktop allows to receive inbound calls and make outdial calls through the internet.

ⓘ When working outside the corporate location, make sure to update your current address in the Emergency service. [See help section.](#)

Team

WebexOne_Team_1

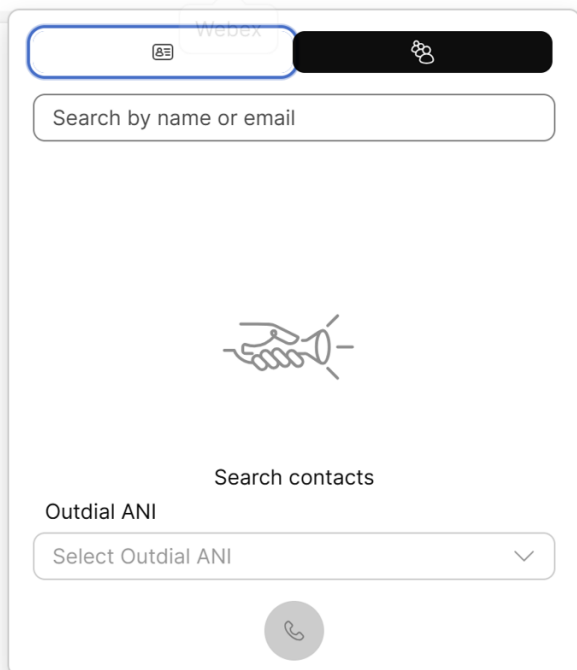
☐ Remember My Credentials

Cancel Submit

- Present task is to dial your cell phone number.
- First, click the Outdial Call option on the top right corner of the desktop.



- You'll notice that the dial pad is missing; the only available option is to search by name, email, or number within the tenant.



Webex

Search by name or email

Search contacts

Outdial ANI

Select Outdial ANI

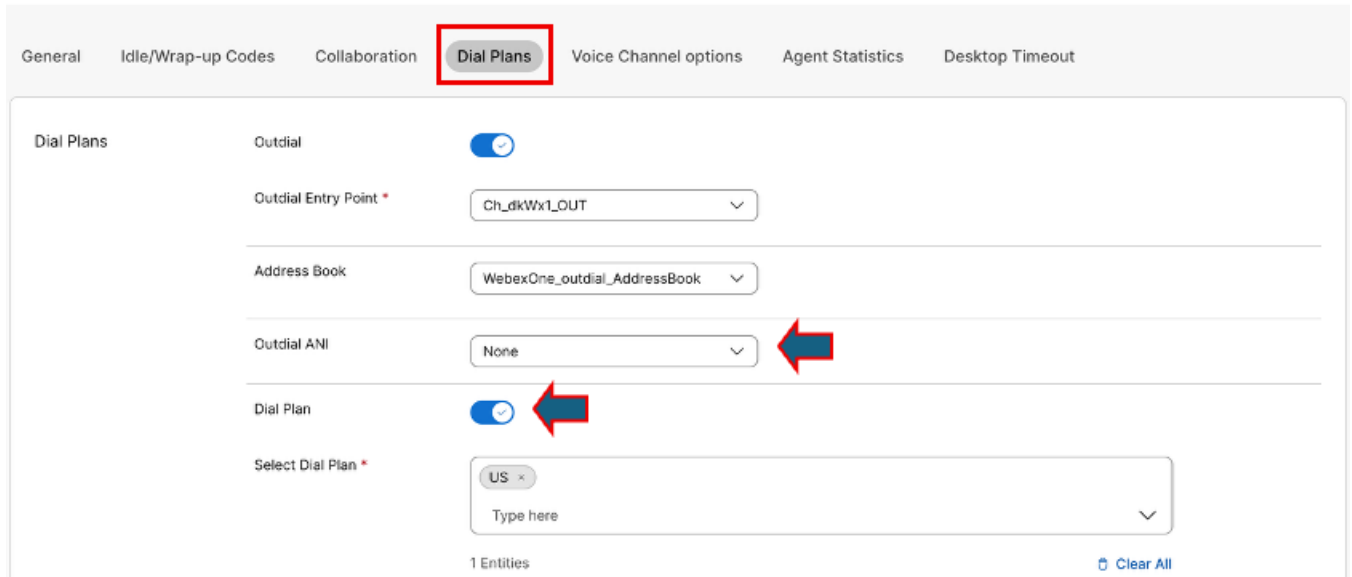
Call icon

- This prevents us from dialing an individual cell phone number directly.
- To fix this, we need to find where the dial pad setting is controlled.
- Since this is an agent desktop function, we'll check the agent's desktop profile and the dial plan where we enabled the outdial option.

- In Control Hub, go back to the Desktop Profile section.
- Select the profile that's mapped to the agent you are working with.
- Navigate to the Dial Plans tab.
- Enable the dial plan functionality and select US as the dial plan. Then, click Save.

dkWx1lab3DeskProfile

ID: a07e4707-67c0-43c4-902c-dbcf2f8285af • Last Modified: August 23, 2025 09:13 AM

🔒  Active


General Idle/Wrap-up Codes Collaboration **Dial Plans** Voice Channel options Agent Statistics Desktop Timeout

Dial Plans

Outdial 

Outdial Entry Point *

Address Book

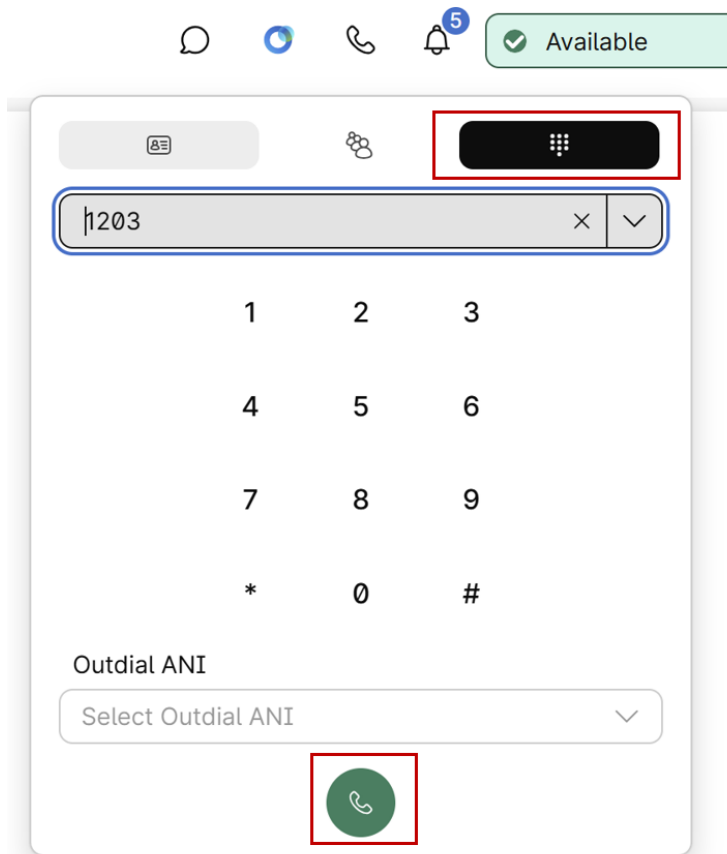
Outdial ANI

Dial Plan 

Select Dial Plan *

1 Entities [Clear All](#)

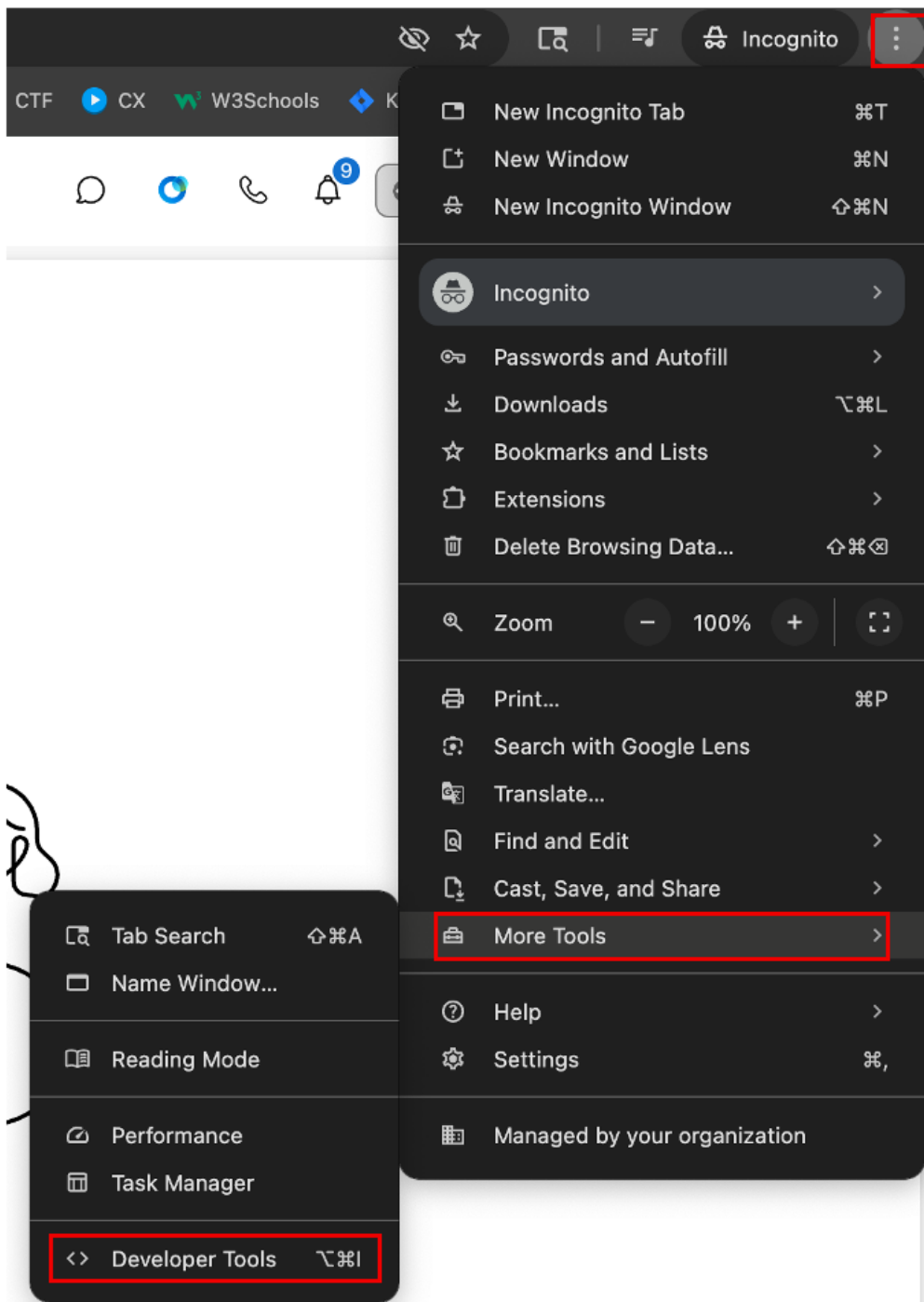
- Refresh the Agent Desktop application browser.
- Click the Outdial option again.
- You should now see the number pad pop up, allowing you to punch in numbers.
- Enter your cell phone number. You may add a "1" before the number, or it will work without it
- Click the Dial button.



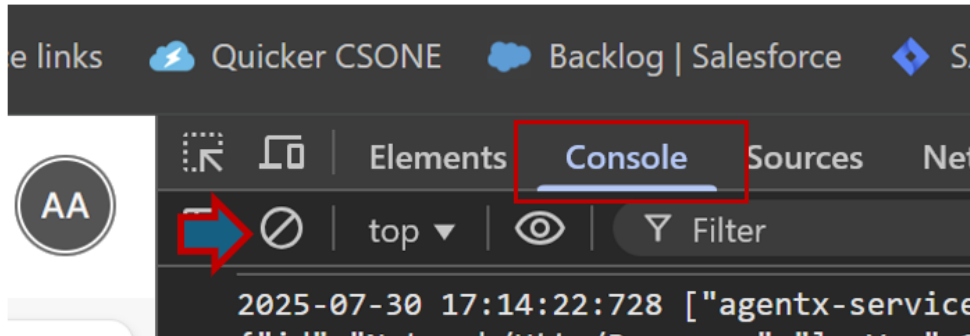
- You should ideally see an agent-initiated call to the cell phone number, but nothing happens.

1.4.4 Section 3 : Troubleshoot Outdial Failure

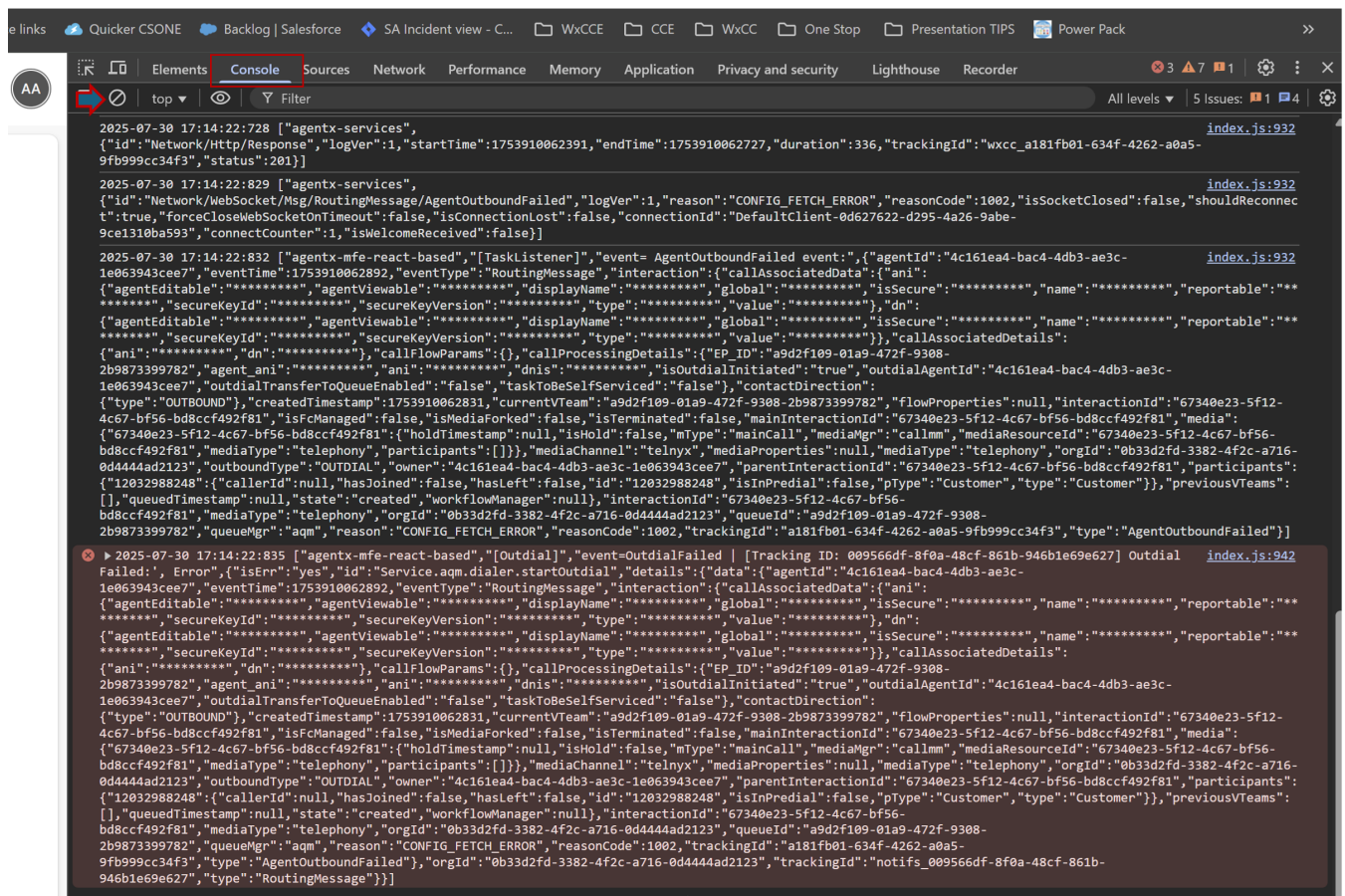
- Let's troubleshoot to see why this is the case.
- To figure this out, bring up the browser developer tool (Windows Shortcut: Press F12 Key)



- Once the developer tool is up, ensure that it's on the "Console" tab and clear the console logs by selecting the "Clear Console" button



- Using the dial pad, dial the cell phone number again.
- As soon as the call fails, you should see a red error message in the console logs with the message **event=OutdialFailed**



- Now, let's look closer into the failure message and figure out what the issue might be.
- For ease, one can copy the error message into a Notepad or Notepad++ application.
- Search for "error," and at the bottom of the error message, you will notice there is a fetch error on "**Config**" - "**Config_fetch_error**." The exact config it's talking about is "**queuemgr**" which basically means queue.

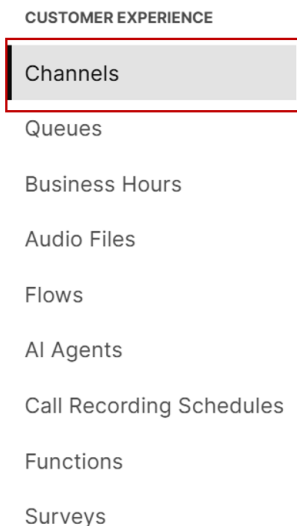
```

2025-07-30 17:14:22:835 [{"agentx-mfe-react-based", "[Outdial]", "event=OutdialFailed | [Tracking ID: 009566df-8f0a-48cf-861b-9461e6e627] Outdial index.js:942
Failed: ", "Error", {"isErr": "yes", "id": "Service.agm.dialer.startOutdial", "details": {"data": {"agentId": "4c161ea4-bac4-4db3-ae3c-
1e063943cee7", "eventTime": 1753910062892, "eventType": "RoutingMessage", "interaction": {"callAssociatedData": {"ani":
{"agentEditable": "*****", "agentViewable": "*****", "displayName": "*****", "global": "*****", "isSecure": "*****", "name": "*****", "reportable": "*****", "secureKeyId": "*****", "secureKeyVersion": "*****", "type": "*****", "value": "*****", "dn":
{"agentEditable": "*****", "agentViewable": "*****", "displayName": "*****", "global": "*****", "isSecure": "*****", "name": "*****", "reportable": "*****", "secureKeyId": "*****", "secureKeyVersion": "*****", "type": "*****", "value": "*****", "dn":
*****", "secureKeyId": "*****", "secureKeyVersion": "*****", "type": "*****", "value": "*****", "dn":
{"ani": "*****", "dn": "*****", "callFlowParams": {}, "callProcessingDetails": {"EP_ID": "a9d2f109-01a9-472f-9308-2b9873399782", "flowProperties": null, "interactionId": "67340e23-5f12-4c67-bf56-bd8ccf492f81", "isFcmManaged": false, "isMediaForked": false, "isTerminated": false, "mainInteractionId": "67340e23-5f12-4c67-bf56-bd8ccf492f81", "media":
{"67340e23-5f12-4c67-bf56-bd8ccf492f81": {"holdTimestamp": null, "isHold": false, "mType": "mainCall", "mediaMgr": "callmm", "mediaResourceId": "67340e23-5f12-4c67-bf56-bd8ccf492f81", "mediaType": "telephony", "participants": []}}, "mediaChannel": "telnyx", "mediaProperties": null, "mediaType": "telephony", "orgId": "0b33d2fd-3382-4f2c-a716-0d444ad2123", "outboundType": "OUTDIAL", "owner": "4c161ea4-bac4-4db3-ae3c-1e063943cee7", "parentInteractionId": "67340e23-5f12-4c67-bf56-bd8ccf492f81", "participants":
{"12032988248": {"callerId": null, "hasJoined": false, "hasLeft": false, "id": "12032988248", "isInPredial": false, "pType": "Customer", "type": "Customer"}}, "previousVTeams":
[], "queuedTimestamp": null, "state": "created", "workFlowManager": null, "interactionId": "67340e23-5f12-4c67-bf56-bd8ccf492f81", "mediaType": "telephony", "orgId": "0b33d2fd-3382-4f2c-a716-0d444ad2123", "queueId": "a9d2f109-01a9-472f-9308-2b9873399782", "queueMgr": "aqm", "reason": "CONFIG_FETCH_ERROR", "reasonCode": 1002, "trackingId": "a181fb01-634f-4262-a0a5-9fb99cc34f3", "type": "AgentOutboundFailed", "orgId": "0b33d2fd-3382-4f2c-a716-0d444ad2123", "trackingId": "notifs_009566df-8f0a-48cf-861b-9461e6e627", "type": "RoutingMessage"}]}

```

"queueMgr": "aqm", "reason": "CONFIG_FETCH_ERROR", "reasonCode": 1002,

- This overall means the system is not able to fetch the team details from the queue perspective where agent resides.
- In WxCC, a queue is always mapped to an entry point via routing flows, so let's go back to the Entry Point for outdial and check the configuration again.
- Via Control Hub, under "Customer Experience," go back via "Channel" to the Outdial Entry Point that was configured.



- Under "Entry Point Settings," you will notice that there is no routing flow mapped.
- From the dropdown, select the flow "**WebexOne_OutdialUser[num]_Flow**" and fill in these fields:
- Music on hold: "defaultmusic_on_hold"
- Version label: Latest
- Outdial Queue: Select the queue that was created in step 1 of your initial setup.

Entry point settings

Service level threshold ⓘ * Seconds

Timezone (Business hours only) * ▼

Routing flow ▼

Music on hold * ▼ 

Version label * ▼ 

The flow version may impact the caller's experience and the variables available to override.

Outdial queue * ▼ 

- Save the settings.
- Refresh the Agent Desktop browser and perform the outdial to the cell phone number; the call should now be successful.
- If you observe the browser debug console logs, you should see a message that will clearly show case its fetching the config and have the team details via the queue ID.

1.4.5 Section 4 : Custom Outdial ANI

- In many cases, business requirements dictate that the Outdial ANI displayed on customer devices should be set to a specific toll-free or departmental number.
- Here, the outdial ANI noticed on the cell phone is **"*+19842906065*"** which is the default configuration set on the tenant level.
- To, review the Tenant-Level Outdial ANI setting in tenant Settings navigate to the Voice tab and note the existing Outdial ANI which is **"*+19842906065*"**.

TENANT SETTINGS

General

Security

Voice 

Digital

Desktop

Integrations

Bulk Operations

Addons

Call Settings

Short Call Threshold seconds

Sudden Disconnect Threshold seconds

Default Outdial ANI ▼

Record all calls ☒

- To change to a custom ANI, administrator can create there own outdial ANI.
- For ease here outdial ANI has already been pre-configured and to review in Desktop Experience, go to Outdial ANI settings.

DESKTOP EXPERIENCE

AI Assistant

Multimedia Profiles

Outdial ANI

Desktop Layouts

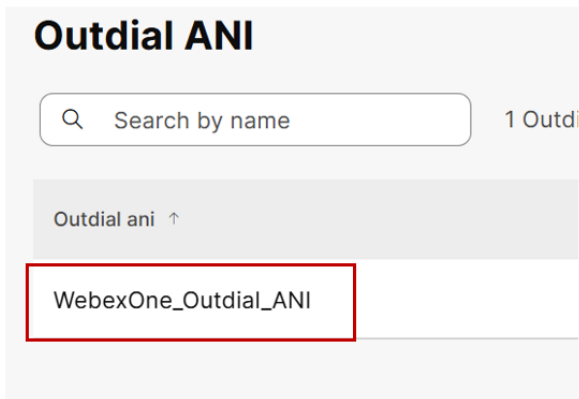
Dial Plans

Address Books

Desktop Profiles

Idle/Wrap-up Codes

- Select **WebexOne_Outdial_ANI**.



- Confirm that it is mapped to the number "**+19842906070**".

General

Name *

WebexOne_Outdial_ANI

Description

Enter a description

Referenced by

There are no references available.

Reference list

Entry List

Number	Name	Contact Number	Actions
1	WebexOneOutdialANI	+19842990206	

Add More

- A custom Outdial ANI allows an agent to select the ANI on the desktop during an outdial call, provided the agent's desktop profile is mapped to the new Outdial ANI.
- To check in Desktop Experience, open the configured Agent Desktop Profile.
- Navigate to the Dial Plans section.
- In the Outdial ANI field, select **WebexOne_Outdial_ANI** and save your changes.

DESKTOP EXPERIENCE

AI Assistant

Multimedia Profiles

Outdial ANI

Desktop Layouts

Dial Plans

Address Books

Desktop Profiles

Idle/Wrap-up Codes

WebexOne_AgentProfile

ID: 9dc1bddc-207a-448f-91c3-593cd5a98338 • Last Modified: August 12, 2025 14:09 PM

General

Idle/Wrap-up Codes

Collaboration

Dial Plans

Voice Channel options

Agent Statistics

Dial Plans

Outdial

Outdial Entry Point *

Address Book

Outdial ANI

Dial Plan

Select Dial Plan *

WebexOne_OutdialUser1_EP

WebexOne_outdial_AddressBook

WebexOne_Outdial_ANI

US

Type here

1 Entities

- Now, Perform an Outdial Call Using the Custom ANI.
- Log in or refresh the agent desktop.
- Initiate an outdial call.
- Verify on the recipient's device that the displayed ANI is the custom number.

Congratulations !! on completing this exercise!

You've not only set up the outdial feature from scratch but also learned how to identify and fix common errors, ensuring your deployments are both functional and reliable.

1.5 Task 4 - Exploring WebRTC data and statistics

Please use the following credentials to connect to Control Hub and configure Webex Contact Center:

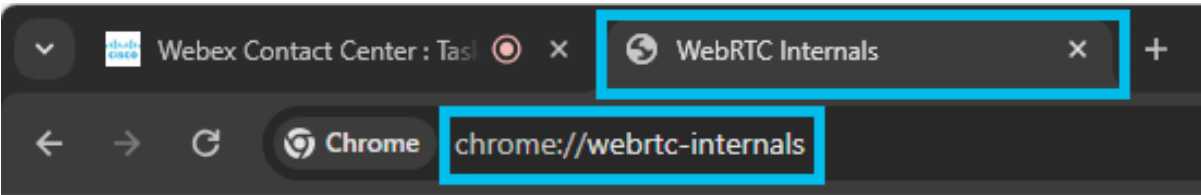
Control Hub URL	https://admin.webex.com
Username	labuserID@wx1.wbx.ai (where ID is your assigned pod number; this ID will be provided by your proctor)
Password	webexONE1!



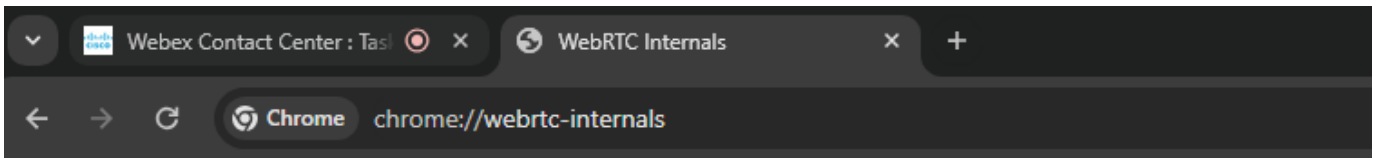
This task showcases available WebRTC call data and statistics using built-in browser tools and simulates missing audio during the call.

1.5.1 Section 1 - Exploring WebRTC internals

- Set your state to **'Available'** on the Agent Desktop.
- Make an inbound call and answer via Agent Desktop when the call is presented.
- Open a new tab in **Google Chrome** web browser.
- Navigate to **chrome://webrtc-internals**.



- In the **WebRTC-internals** tab select the active WebRTC session part (**desktop.wxcc-us1.cisco.com** in its name).



- Create a WebRTC-Internals dump
- Create diagnostic audio recordings
- Create diagnostic packet recordings



• The following information can be seen:

- On the left side **(1)**, you'll find a record of the PeerConnection API activities. The RTCPeerConnection is the central interface in the WebRTC API. It represents the connection between the local and remote peer and provides all the functions and events necessary to establish the connection (more details can be found at <https://webrtc.org/getting-started/overview>). These traces **(1)** display interactions with the RTCPeerConnection object and parameters used.
- On the right side **(2)**, you'll find the metrics collected via the getStats API about the current WebRTC connection.
- On the bottom **(3)**, you'll find graphs produced by utilizing the getStats API **(2)** about the current WebRTC connection.

<https://desktop.wxcc-us1.cisco.com/>, { iceServers: [], iceTransportPolicy: all, bundlePolicy: max-compat, rtcpMuxPolicy: require, iceCandidatePoolSize: 0 },

1

```

ICE connection state: new => completed
Connection state: new => connected
Signaling state: new => stable
ICE Candidate pair: 10.24.139.247:49870 <=> 23.89.1.162:22744
► ICE candidate grid
Time                Event
4/17/2024, 10:13:52 AM ► setRemoteDescription (type: "offer", 2 sections)
4/17/2024, 10:13:52 AM ► setLocalDescription (type: "answer", 2 sections)
4/17/2024, 10:13:52 AM ► signalingstatechange
4/17/2024, 10:13:52 AM ► iceconnectionstatechange
4/17/2024, 10:13:52 AM ► connectionstatechange

```

2

Stats Tables

Filter statistics by type including

```

► media-playout (kind=audio, id=AP)
► certificate (id=CFE0AA:E8:8D:CE:FA:A0:0E:63:4B:6C:16:C3:D0:AD:C3:7B:2B:A7:EE)
► certificate (id=CFFD:14:97:87:1A:41:7C:4C:14:02:AD:23:19:54:FE:23:E6:4F:61:AD:39:B1:47:C6:A8:AB:39:4E:35:FC:71:F8)
► codec (mimeType=audio/PCMU, payloadType=0, id=CIT01_0)
► codec (mimeType=audio/PCMU, payloadType=0, id=COT01_0)
► candidate-pair (state=succeeded, id=CPyDlSaDsy_T5aV8Gj8)
► local-candidate (candidateType=host, id=10YJvKsp)
► local-candidate (candidateType=host, tcpType=active, id=I9OkJn/oQ)
► local-candidate (candidateType=host, tcpType=active, id=ILUW6L1VJ)
► inbound-rtp (kind=audio, mid=0, ssrc=577774253, [codec]=PCMU (0), id=IT01A577774253)
► remote-candidate (candidateType=host, id=IT5aV8Gj8)
► local-candidate (candidateType=host, tcpType=active, id=IcCmyDGdn)
► local-candidate (candidateType=host, id=id4Nah5bb)
► local-candidate (candidateType=host, tcpType=active, id=Ih75e1cck)
► local-candidate (candidateType=host, id=IhDILG2ss)
► local-candidate (candidateType=host, id=lyDlSaDsy)
► outbound-rtp (kind=audio, mid=0, ssrc=3932673161, [codec]=PCMU (0), id=OT01A3932673161)
► peer-connection (id=P)
► remote-inbound-rtp (kind=audio, ssrc=3932673161, id=RIA3932673161)
► remote-outbound-rtp (kind=audio, ssrc=577774253, id=ROA577774253)
► media-source (kind=audio, id=SA1)
► transport (iceState=connected, dtlsState=connected, id=T01)

```

3

Filter statistics graphs by type including

```

► Stats graphs for media-playout (kind=audio, id=AP)
► Stats graphs for candidate-pair (state=succeeded, id=CPyDlSaDsy_T5aV8Gj8)
► Stats graphs for inbound-rtp (kind=audio, mid=0, ssrc=577774253, [codec]=PCMU (0), id=IT01A577774253)
► Stats graphs for outbound-rtp (kind=audio, mid=0, ssrc=3932673161, [codec]=PCMU (0), id=OT01A3932673161)
► Stats graphs for peer-connection (id=P)
► Stats graphs for remote-inbound-rtp (kind=audio, ssrc=3932673161, id=RIA3932673161)
► Stats graphs for remote-outbound-rtp (kind=audio, ssrc=577774253, id=ROA577774253)
► Stats graphs for media-source (kind=audio, id=SA1)
► Stats graphs for transport (iceState=connected, dtlsState=connected, id=T01)

```

- Local and remote IPs and ports in use for this particular WebRTC connection can be seen under traces of the PeerConnection API calls.

ICE connection state: new => completed

Connection state: new => connected

Signaling state: new => stable

ICE Candidate pair: 10.24.139.247:49870 <=> 23.89.1.162:22744

► ICE candidate grid

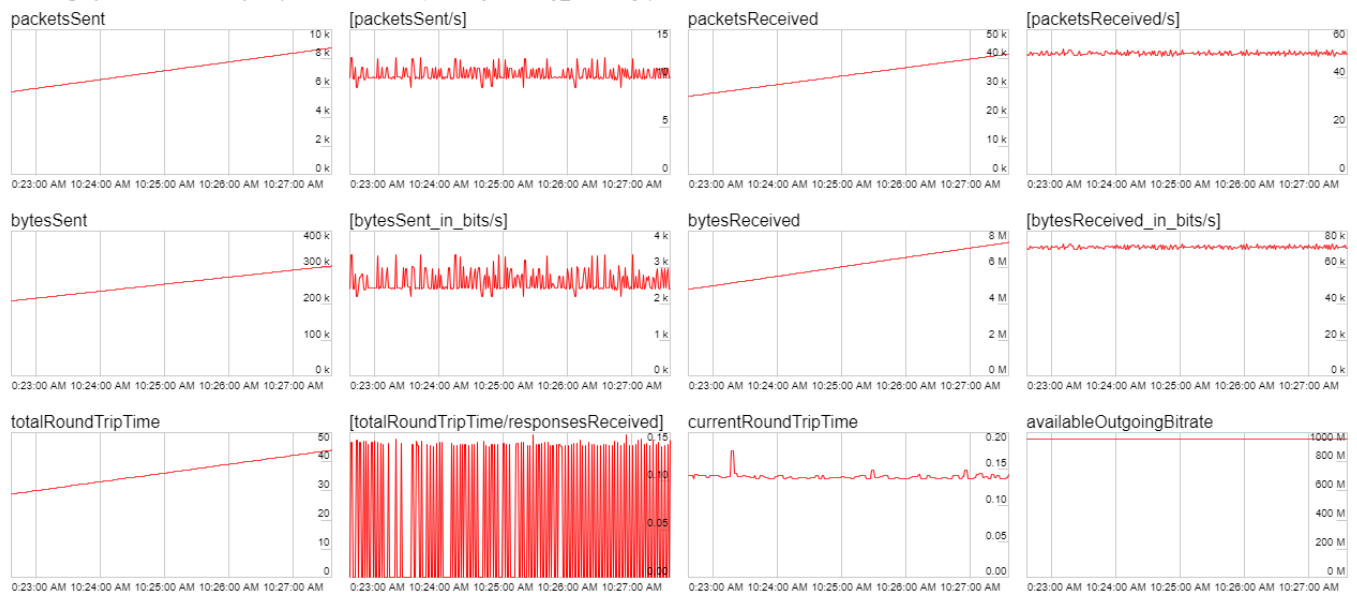
Time	Event
4/17/2024, 10:13:52 AM	► setRemoteDescription (type: "offer", 2 sections)
4/17/2024, 10:13:52 AM	► setLocalDescription (type: "answer", 2 sections)
4/17/2024, 10:13:52 AM	► signalingstatechange
4/17/2024, 10:13:52 AM	► iceconnectionstatechange
4/17/2024, 10:13:52 AM	► connectionstatechange

- Various live call statistics (i.e. sent packets, received packets, latency, dropped packets etc.) related graphs can be seen at the bottom.

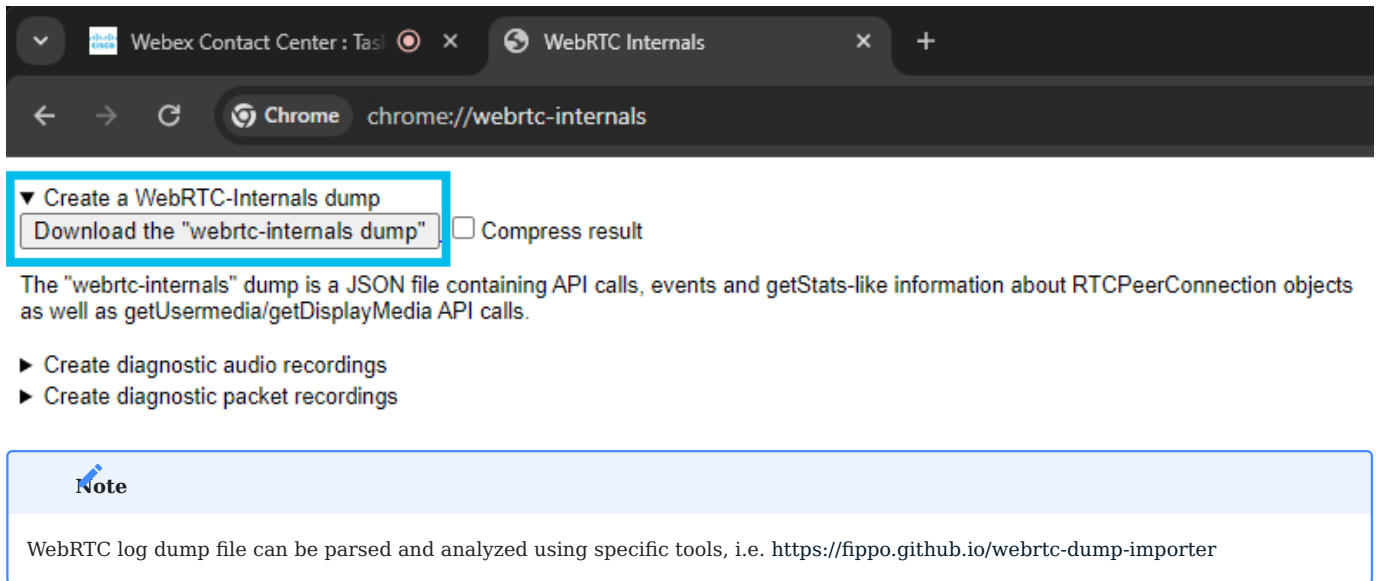
Filter statistics graphs by type including

► Stats graphs for media-playout (kind=audio, id=AP)

▼ Stats graphs for candidate-pair (state=succeeded, id=CPyDI\$SaDsy_T5aV8Gj8)



- WebRTC log dump can be retrieved by clicking '**Create a WebRTC-Internals dump**' and clicking '**Download the "webrtc-internals dump"**' option.



▼ Create a WebRTC-Internals dump

Download the "webrtc-internals dump" ☐ Compress result

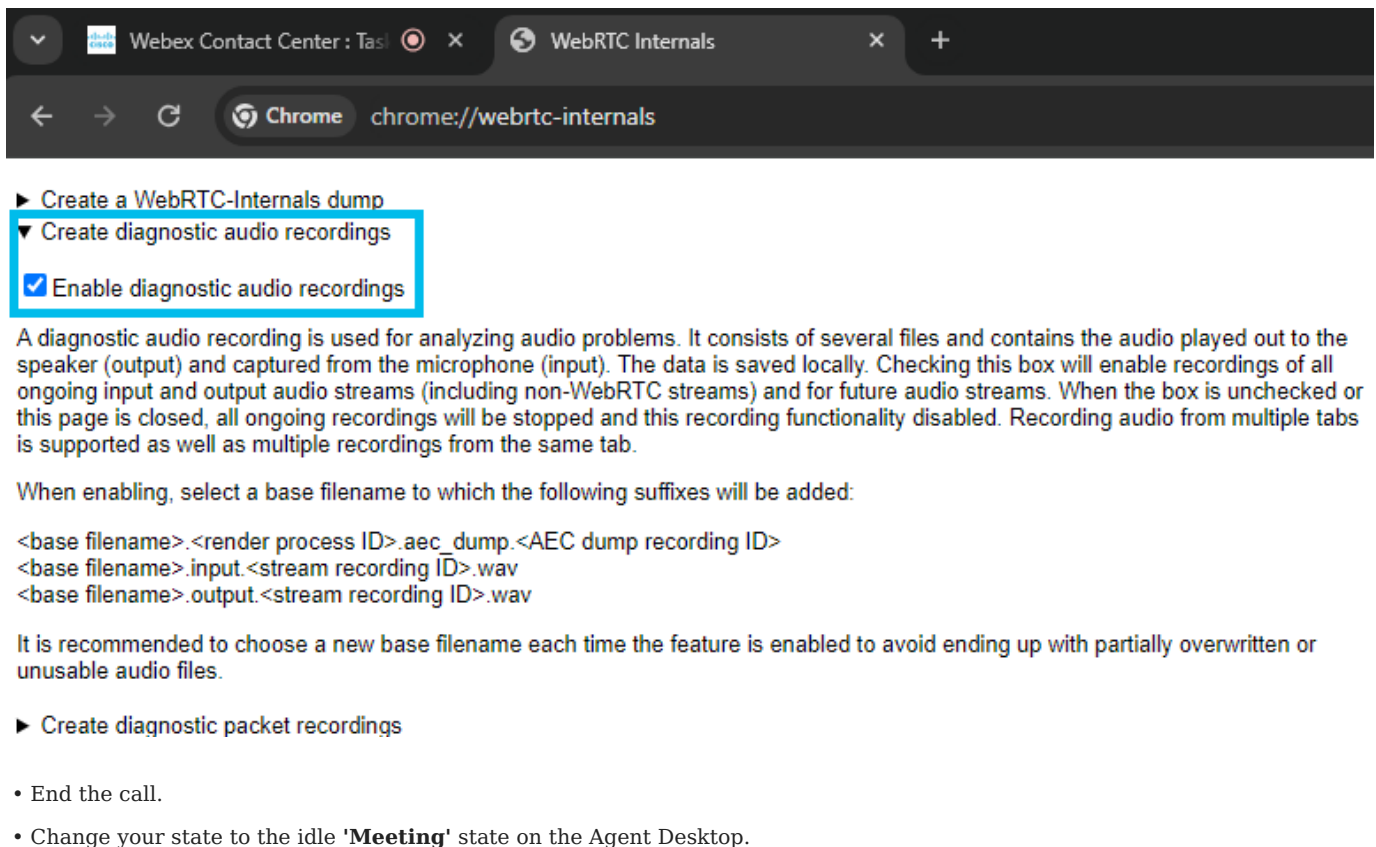
The "webrtc-internals" dump is a JSON file containing API calls, events and getStats-like information about RTCPeerConnection objects as well as getUsermedia/getDisplayMedia API calls.

- ▶ Create diagnostic audio recordings
- ▶ Create diagnostic packet recordings

Note

WebRTC log dump file can be parsed and analyzed using specific tools, i.e. <https://fippo.github.io/webrtc-dump-importer>

- WebRTC audio can also be captured for diagnostic purposes.



▶ Create a WebRTC-Internals dump

▼ Create diagnostic audio recordings

☒ Enable diagnostic audio recordings

A diagnostic audio recording is used for analyzing audio problems. It consists of several files and contains the audio played out to the speaker (output) and captured from the microphone (input). The data is saved locally. Checking this box will enable recordings of all ongoing input and output audio streams (including non-WebRTC streams) and for future audio streams. When the box is unchecked or this page is closed, all ongoing recordings will be stopped and this recording functionality disabled. Recording audio from multiple tabs is supported as well as multiple recordings from the same tab.

When enabling, select a base filename to which the following suffixes will be added:

```
<base filename>.<render process ID>.aec_dump.<AEC dump recording ID>
<base filename>.input.<stream recording ID>.wav
<base filename>.output.<stream recording ID>.wav
```

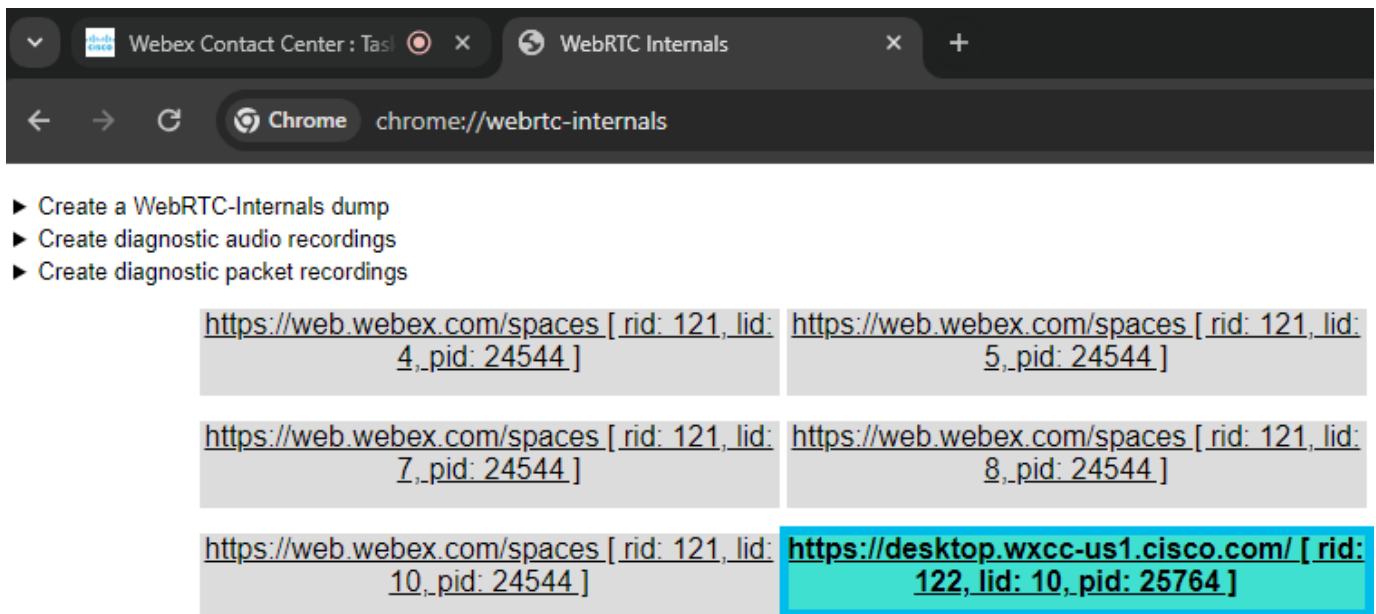
It is recommended to choose a new base filename each time the feature is enabled to avoid ending up with partially overwritten or unusable audio files.

- ▶ Create diagnostic packet recordings

- End the call.
- Change your state to the idle '**Meeting**' state on the Agent Desktop.

1.5.2 Section 2 - Investigating missing audio

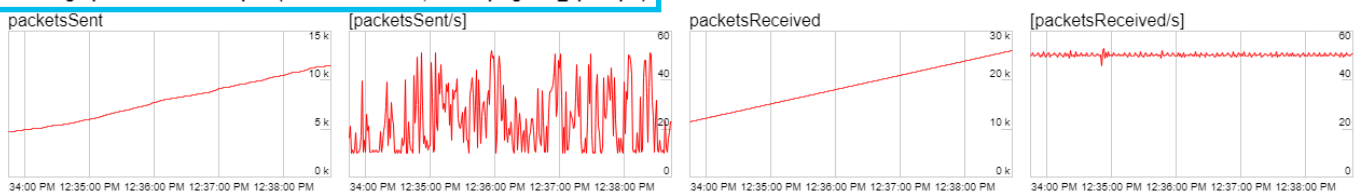
- Download **task3s2.zip** file from <https://github.com/asuchank/clus24/raw/main/task3s2.zip> and extract it.
- Make a new call to **+19842906065** using WebRTC via Agent Desktop and press **0** (music should be playing).
- Navigate to **chrome://webrtc-internals** in a new tab.
- In the **WebRTC-internals** tab select the active WebRTC session part (it will have **desktop.wxcc-us1.cisco.com** in its name).



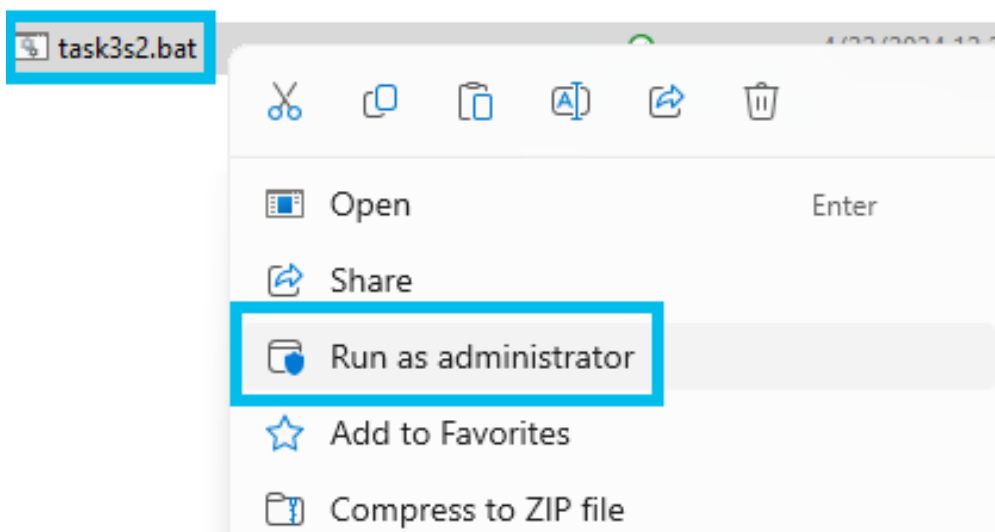
- At the bottom of the **WebRTC-internals** diagnostic page expand bolded text to uncover graphs for an ongoing session.

Filter statistics graphs by type including

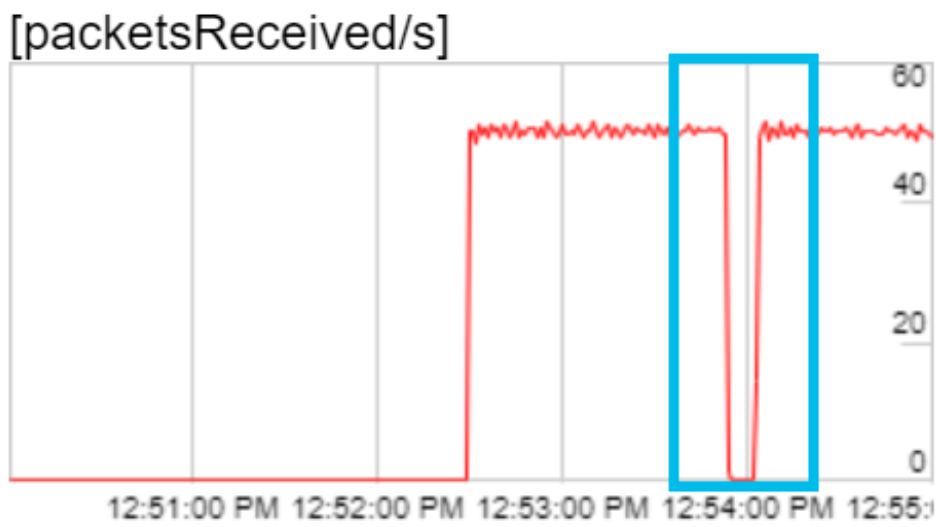
- ▶ Stats graphs for media-playout (kind=audio, id=AP)
- ▶ Stats graphs for candidate-pair (state=in-progress, id=CP6i8FYXMa_epa/oqzv)
- ▼ Stats graphs for candidate-pair (state=succeeded, id=CPq5eg+09r_epa/oqzv)



- Right-click on the **task3s2.bat** file and choose 'Run as administrator' (audio will be lost for 10 seconds).



- Observe **packetsReceived/s** go down to zero on the expanded graph from before.



- Based on the observation above - missing audio was caused by the lack of media packets.
- End the call.
- Congratulations, you have completed this task!

1.6 Conclusion

1.6.1 Create the Skill-Based Queue

 **Video Walkthrough: Create Skill-Based Queue**

Tip: Click the  on the video player and select **Picture-in-Picture** to keep the video floating on screen while performing lab steps in Control Hub.

1. In the **Contact Center** navigation pane, select **Queues** under the **Customer Experience** section.
2. Click the **Create Queue** button.
3. Configure the general queue settings using the table below:

Field Name	Value / Setting	Description
Queue Name	<code>WebexOne_SBR_Queue_[name]</code>	
Contact Direction	Inbound Queue	
Channel Type	Telephony	

1. Scroll to **Contact Routing Settings**, enable **Skill-Based Routing**, and configure the following:
 - **Skill Assignment Type:** Select **Assign skills in flows**.
 - **Agent Assignment:** Select **Teams**.
 - **Routing Pattern:** Select **Longest Available**.
2. Under **Call Distribution**, create a new group and add your designated team: `WebexOne_Team_[num]`.
3. Scroll to **Advanced Settings** and verify the following parameters:
 - **Service Level Threshold:** `30` seconds
 - **Maximum Time in Queue:** `30` seconds
 - **Default Music in Queue:** `defaultmusic_on_hold.wav`
4. Click **Save** to finalize and create the queue.